

# OntoRisk: An Ontology-Guided Event-Centric and Evidence-Constrained Framework for AI Risk Knowledge Graph Construction from Multi-Source Incident Reports

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## ABSTRACT

Real-world AI incidents are documented in thousands of fragmented, multi-source reports that resist auditable analysis: document-centric extraction scatters one incident across sources, and unconstrained large language model (LLM) extraction emits statements with unverifiable provenance. Here we present **OntoRisk**, which makes LLM-based knowledge graph construction aggregatable and auditable by binding extraction to an event-centric risk ontology with mandatory evidence. We extend the AI Risk Ontology (AIRO) into a 61-class AI Risk Incident Ontology centered on *AIRiskIncident*, reified stakeholder *RoleAssignment*, and a five-slot propagation chain (*RiskSource*, *Risk*, *Consequence*, *Impact*, *AffectedActor*). OntoRisk aggregates cross-document reports into event clusters and acquires knowledge in three evidence-constrained layers: explicit entity extraction, risk-chain construction, and ontology-guided inference of abstracted incident attributes, with every statement retaining its source span, extraction mode, and confidence. On 2,034 event clusters aggregated from 6,124 documents (1998–2025), a 100-cluster gold standard built under a verification-oriented annotation protocol yields an entity F1 of 92.3%, risk-chain accuracy of 93.3%, and ontology-guided inference accuracy of 96.9%. LLM-judged evidence support reaches 84.4% across the evaluation graph's 12,361 semantic edges, and a manual audit of inference-layer statements yields an evidence support rate of 98.3%. Ablations reveal a coverage–grounding tradeoff: relative to the unconstrained variant, the full system emits 3.7 times as many semantic edges with an 8.2 percentage-point lower LLM-judged support rate. Full-corpus analysis traces the documented shift from technical-safety failures toward generative-AI-enabled misuse, and surfaces recurring propagation pathways and stakeholder response patterns. Event-centric, evidence-constrained knowledge graphs can therefore serve as an auditable foundation for AI risk governance.

## 1. Introduction

Artificial intelligence (AI) systems now operate at scale in high-stakes domains, and their failures—privacy violations, discriminatory outcomes, safety failures, governance controversies—leave a growing documentary trail in news reports and public databases such as the AI Incident Database and AIAAIC [1, 2]. For regulators, auditors, and researchers, these records are the primary empirical basis for understanding how AI risks arise, propagate, and are responded to, yet they resist systematic use: a single incident is covered by multiple documents reporting complementary fragments at different granularity, and no established method reconstructs this fragmented coverage into an auditable, event-level representation. This work addresses the task of constructing event-centric AI risk knowledge graphs (KGs) from multi-source incident reports, recovering for each documented incident its stakeholders, involved systems, risk propagation structure, and governance responses, with every recovered claim traceable to its sources.

This task is challenging for three reasons. First, fragmentation. The same incident arrives as partially overlapping documents with inconsistent vocabulary,

and document-centric knowledge graph construction processes each report independently [3–5]; entities therefore multiply, complementary facts remain scattered, and cross-document conflicts go undetected. Second, verifiability. Large language models (LLMs) have become the dominant tool for open-domain entity, relation, and event extraction [6, 7], but unconstrained generation produces statements whose provenance is not recorded, so downstream users cannot distinguish text-grounded facts from unsupported generations; governance analysis cannot rest on such statements, because responsibility attribution is meaningful only when each claim can be traced to evidence. Anchor-constrained extraction with per-triplet provenance mitigates this within single documents [8], but provenance has not been integrated with event-level aggregation or ontology-guided constraints. Third, semantics. Existing AI risk ontologies such as the AI Risk Ontology (AIRO) [9] provide a principled vocabulary aligned with the European Union AI Act, but they model static risk categories for compliance-oriented description rather than incident events, with no first-class support for evidence provenance, incident-specific stakeholder roles, or risk propagation, and thus cannot serve as operational schemas that constrain extraction over incident reports.

We present **OntoRisk**, a framework built on the observation that aggregation and auditability need not trade off: binding LLM-based extraction to an event-centric risk

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ontology with mandatory evidence yields graphs that are consolidated across sources and, at the same time, verifiable statement by statement. We extend AIRO into an AI Risk Incident Ontology centered on *AI Risk Incident* as a semantic hub, reified *RoleAssignment* for incident-specific stakeholder contextualization, and a five-slot risk propagation chain (*RiskSource*, *Risk*, *Consequence*, *Impact*, *AffectedActor*) that makes event-level risk causality a first-class structure. Evidence spans, extraction modes, and confidence scores are attached to every knowledge statement, so the graph records not only what is known but also how each item is known. On this schema, OntoRisk aggregates cross-document reports into event clusters and acquires knowledge in three constrained layers, explicit entity extraction, risk-chain construction, and ontology-guided inference, each bounded by ontology types and grounded in the evidence produced by the layers below it.

We evaluate OntoRisk on a benchmark of 2,034 event clusters aggregated from 6,124 incident documents spanning 1998 to 2025. Against a human-annotated gold standard of 100 clusters, the framework attains high accuracy across all three extraction layers, and ablations isolate the role of each component, revealing a tradeoff between structural coverage and per-statement evidence support. Applying the complete pipeline to the full corpus characterizes the documented incident landscape at scale, tracing the shift from early technical-safety failures toward generative-AI-enabled misuse, recurring propagation pathways, and patterns of stakeholder and regulatory response.

The main contributions of this work are summarized as follows:

- An AI Risk Incident Ontology that extends AIRO with event-centric incident modeling, reified stakeholder role assignment, five-slot risk propagation chains, governance response representation, and evidence provenance as first-class concepts.
- OntoRisk, an ontology-guided, event-centric, and evidence-constrained framework that integrates cross-document event aggregation with three-layer knowledge acquisition for AI risk knowledge graph construction.
- A benchmark of 2,034 event clusters built from multi-source incident reports, together with an ontology-guided annotation protocol and a 100-cluster gold standard supporting multi-layer human evaluation.
- Experiments and ablation studies quantifying extraction accuracy, evidence support, and the complementary roles of the four framework components, together with full-corpus application analyses of the documented AI risk incident landscape.

## 2. Related Work

### 2.1. Knowledge Graph Construction

Knowledge graph construction aims to transform unstructured textual data into structured semantic representations. Early studies relied on pipeline architectures composed of named entity recognition, relation extraction, entity linking, and ontology alignment modules [4], which performed well in domain-specific settings but suffered from error propagation and schema dependency. More recent approaches build on pretrained language models and large language models to improve open-domain extraction and semantic generalization [6], and LLM-based frameworks have proved effective for structured triple generation, schema-aware extraction, and knowledge completion [10]. A related line grounds LLM extraction in the source text itself: the Anchor-Extraction-Verification-Supplement framework constrains triplet generation to anchors discovered in the document and tracks per-triplet provenance, substantially reducing hallucination on general-domain benchmarks [8], establishing that grounding constraints are a primary mechanism for suppressing unsupported generations. OntoRisk differs in the unit and the source of constraint: it grounds statements at the event level across documents, and it derives its constrained vocabulary from a domain ontology rather than from surface anchors, so that grounding and semantic typing are enforced jointly.

Despite these advances, most existing KG construction methods are designed around sentence-level or document-level extraction, focusing on local entity-relation prediction and rarely modeling event-level semantic structures across multiple documents [5]. Ontology constraints are often used only as auxiliary schemas rather than integrated into extraction and aggregation [11], so existing methods may produce semantically inconsistent or weakly grounded knowledge graphs in complex real-world scenarios.

### 2.2. Event-Centric and Cross-Document Information Extraction

Event extraction has been widely studied for identifying structured event representations from text, including triggers, arguments, and semantic roles. Early work focused on supervised feature-engineering approaches [12]; later studies introduced neural architectures and joint extraction models, and recent pretrained language model approaches further improved open-domain extraction [7].

To capture event information distributed across multiple sources, cross-document event extraction and event coreference resolution have attracted increasing attention, aiming to identify and aggregate event mentions referring to the same real-world incident across heterogeneous documents [13]. More recent studies explored multi-document event extraction frameworks that jointly perform event alignment and information aggregation [5]. Among them, the CDEE pipeline is the closest to our aggregation stage; it chains event extraction, coreference resolution, entity normalization, role normalization, and entity-role resolution, and reports about 72% F1 end-to-end on a

general-domain benchmark [5]. These approaches improve global event understanding by integrating complementary information from multiple reports, but their alignment decisions are not tied to a domain schema, their outputs do not carry evidence spans, and governance-related semantics are out of scope. Current cross-document extraction methods thus focus mainly on event clustering and alignment, with limited attention to ontology-guided semantic constraints, evidence-aware knowledge representation, and the stakeholder and risk propagation structures central to AI risk incidents.

### 2.3. AI Risk Ontologies and Governance Modeling

With the increasing deployment of AI systems in high-impact domains, researchers and policymakers have proposed various frameworks for AI governance and risk management. International initiatives such as the OECD AI Principles [14] and the European Union AI Act [15] establish regulatory foundations for trustworthy and risk-aware AI systems, motivating ontology-based approaches for formalizing AI governance concepts and supporting machine-readable policy analysis. Among existing efforts, the AI Risk Ontology (AIRO) provides a structured semantic framework grounded in the EU AI Act and related governance standards, modeling AI system lifecycle components, stakeholders, and regulatory concepts. Recent studies have further explored the integration of ontologies with LLM-based pipelines, including ontology-based verification of LLM-extracted triplets [10], roadmaps for unifying LLMs and knowledge graphs [16], and compilation of ontologies into executable constraints for LLM agents [17]. Orthogonally, the Semantic Web community maintains general-purpose provenance and reification standards: PROV-O models entities, activities, and derivations [18]; RDF-star attaches statement-level annotations [19]; nanopublications package assertions together with provenance and publication information [20]; and n-ary relation patterns reify role-like relations [21, 22]. These standards specify how provenance should be represented once it exists, but as neutral vocabularies they neither constrain how an extraction system acquires statements from evidence nor operationalize incident semantics; this work reuses them at the representation level (Section 5.2) and contributes the acquisition-side protocol that binds LLM extraction to them. Nevertheless, existing AI risk ontologies mainly focus on static conceptual modeling and compliance-oriented representations, lacking explicit mechanisms for dynamic incidents, cross-document event aggregation, and evidence-grounded knowledge statements, and they are rarely integrated with automated large-scale extraction frameworks for real-world AI risk analysis [9].

### 2.4. AI Risk Taxonomies and Incident Resources

A complementary line of work develops structured taxonomies and repositories for AI risks and incidents. The AI

Incident Database [1] and AIAAIC [2] curate document-level incident records but do not provide event-level semantic structures or evidence-linked knowledge graphs. The MIT AI Risk Repository [23] consolidates risk frameworks into a database with both a causal taxonomy (causes, risks, and impacts) and a domain taxonomy (seven domains and 23 subdomains); NIST AI 600-1 [24] enumerates generative-AI risks and risk sources; and the AI System Threat Vector Taxonomy [25] catalogs threat vectors and affected parties. These efforts offer controlled vocabularies for risk classification, but they are maintained as standalone reference resources rather than as operational ontologies that constrain extraction, aggregation, and inference over incident reports. The five-slot risk propagation chain adopted in this work is conceptually related to causal-chain modeling traditions in safety engineering, such as bow-tie structures linking threats, top events, and consequences; in this correspondence *RiskSource* plays the role of a threat on the prevention side, *Risk* the top event, and *Consequence* and *Impact* the mitigation side, while *AffectedActor* records the stakeholder bearing the final impact, and explicit barrier modeling is out of scope; the present design trades expressive completeness for interpretability and governance traceability. The seven-domain, 23-subdomain classification used for event categorization is adopted verbatim from the repository's domain taxonomy (Section 7.1, Appendix B).

### 2.5. Summary

Existing studies have advanced KG construction, event extraction, and AI governance modeling from different perspectives, but limited work jointly integrates ontology-guided semantic modeling, cross-document event aggregation, and evidence-constrained extraction within a unified framework for AI risk knowledge graph construction. This work addresses the gap by extending AIRO into an event-centric ontology combined with cross-document extraction and evidence-grounded knowledge aggregation.

## 3. Datasets

We construct a benchmark dataset by integrating two public AI incident repositories: the AI Incident Database (AIID) [1] and the AI, Algorithmic and Automation Incidents and Controversies Repository (AIAAIC) [2]. Both collect real-world AI incidents from news articles, organizational reports, and public disclosures, but are organized at the document level and contain many discussion-oriented entries without concrete risk events.

We therefore apply a two-stage refinement. First, we align reports across AIID and AIAAIC by aggregating records referring to the same real-world event into unified clusters based on shared entities, temporal information, and risk characteristics; this instantiates the clustering mechanism of Section 5, which combines semantic embedding similarity with stakeholder and AI-system entity overlap under temporal consistency constraints. The resulting event-aligned benchmark dataset—RiskNet [26]—provides the alignment and multi-dimensional annotation protocol for

**Table 1**

Statistics of the constructed AI risk incident benchmark dataset.

| Statistic                        | Value     |
|----------------------------------|-----------|
| Total source documents           | 6,124     |
| Total event clusters             | 2,034     |
| Number of covered countries      | 42        |
| Time span of collected incidents | 1998–2025 |
| Average documents per cluster    | 3.01      |
| Median documents per cluster     | 2         |
| Clusters with multiple documents | 59.2%     |

this construction, whose full implementation details are reported therein. Second, we filter out discussion-level entries, retaining only incidents with concrete harmful outcomes, governance violations, safety failures, or socially impactful events (e.g., privacy breaches, discriminatory decision-making, misinformation propagation, harmful autonomous behaviors).

Each resulting cluster contains one or more documents describing the same incident from complementary perspectives, with source-level evidence links preserved. Of the 2,034 clusters, 1,965 have a recorded incident year; among these, 1,647 (83.8%) fall in the 2018–2024 window, and 69 clusters lack a recorded year and are excluded from temporal statistics. Cluster sizes are right-skewed: the 25th, 50th, 75th, and 90th percentiles are 1, 2, 3, and 6 documents, and the largest cluster contains 54 documents. Overall, 829 clusters (40.8%) contain a single document and 59.2% aggregate multiple documents. Table 1 summarizes the resulting dataset.

## 4. Problem Formulation

Given a collection of multi-source incident documents, our goal is to construct an event-centric, evidence-grounded, and ontology-aligned AI risk knowledge graph that supports governance auditing, responsibility attribution, and interpretable risk reasoning. Unlike document-level extraction, which processes each report in isolation, this task requires aggregating information distributed across multiple reports of the same incident while preserving semantic provenance and evidence traceability.

### 4.1. Input

Let

$$\mathcal{D} = \{d_1, d_2, \dots, d_n\} \quad (1)$$

denote incident documents from heterogeneous sources, where each document may describe AI systems, stakeholders, governance actions, risk propagation, and societal impacts, and multiple documents may cover the same incident from different perspectives.

### 4.2. Output

The objective is an AI risk knowledge graph

$$\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{S}, \mathcal{P}), \quad (2)$$

where  $\mathcal{E}$  denotes ontology-aligned entities,  $\mathcal{R}$  semantic relations,  $\mathcal{S} = \{s_1, \dots, s_q\}$  evidence-backed knowledge statements, and  $\mathcal{P}$  provenance information covering evidence sources, extraction modes, and confidence. Unlike conventional triple-centric graphs,  $\mathcal{G}$  organizes knowledge around event-centric incident structures and causal risk propagation chains, enabling governance auditing and accountability analysis.

### 4.3. Event Cluster

An AI risk event cluster

$$c_i = \{d_{i1}, d_{i2}, \dots, d_{i,n_i}\} \quad (3)$$

is a set of documents referring to the same real-world AI risk incident, aggregating complementary information into more complete representations of stakeholders, risk evolution, governance responses, and societal consequences.

### 4.4. Extraction Targets

For each event cluster  $c_i$ , the system aims to extract the following four types of structured knowledge.

1. **Entities**  $e \in \mathcal{E}$ , including AI systems, AI models, AI techniques, AI capabilities, stakeholders (developers, providers, deployers, users, regulators), risk-related entities (risk sources, risks, consequences, impacts, affected actors, risk controls), and governance entities (regulations, standards, obligations).
2. **Relations**  $r(e_i, e_j) \in \mathcal{R}$ , describing semantic interactions such as deployment, governance, causality, responsibility, compliance, mitigation, and societal impact.
3. **Risk Propagation Chains**

$$v_k = (RS, RK, C, I, AA), \quad (4)$$

where  $RS$  denotes the risk source,  $RK$  represents the risk,  $C$  corresponds to the consequence,  $I$  denotes the resulting impact, and  $AA$  represents the affected actor. Together, these components form a causal risk propagation chain:

$$RS \xrightarrow{\text{causes}} RK \xrightarrow{\text{leadsTo}} C \xrightarrow{\text{impacts}} I \xrightarrow{\text{affects}} AA. \quad (5)$$

The arrows represent causal semantics reported by the source evidence and do not imply causal certainty beyond the reports; where evidence supports only temporal or associative relations, the statement uses the appropriate non-causal relation type and confidence.

### 4. Evidence Statements

$$s = (h, r, t, \epsilon, \mu, \gamma), \quad (6)$$

where  $h$  and  $t$  denote the head and tail entities,  $r$  the semantic relation,  $\epsilon$  the supporting evidence span extracted from source documents,  $\mu \in \{\text{explicit}, \text{abstracted}, \text{inferred}\}$  the extraction mode, and  $\gamma \in [0, 1]$  the confidence score. Each  $s_i$  is an individual knowledge statement in the statement set  $\mathcal{S}$ .

#### 4.5. Knowledge Categories

To improve reliability and traceability, OntoRisk distinguishes three categories of extracted knowledge:

- **Explicit knowledge:** directly supported by textual evidence from source documents;
- **Abstracted knowledge:** obtained through semantic normalization or structured summarization across multiple evidence spans;
- **Inferred knowledge:** derived through ontology-guided controlled inference, strictly grounded in evidence produced at lower extraction layers.

These categories are recorded per statement through the extraction mode  $\mu$  in the evidence statement tuple: entity-layer statements are marked explicit, risk-chain statements are marked abstracted because slot values are normalized from and summarized across multiple evidence spans, and inference-layer statements are marked inferred.

#### 4.6. Task Decomposition and Logical Dependencies

The construction of  $\mathcal{G}$  decomposes into three logically ordered sub-tasks:

1. **Cross-document event aggregation:** documents in  $\mathcal{D}$  are partitioned into event clusters  $\mathcal{C} = \{c_1, \dots, c_K\}$ , and local evidence packages are merged into Global Event Representations (GERs). This step is the prerequisite for all subsequent processing, as downstream extraction and inference operate over GERs.
2. **Ontology-guided structured extraction:** entities, relations, and risk propagation chains are extracted from each GER under the constraints of the AI Risk Incident Ontology. This step depends on aggregation because ontology slot-filling requires consolidated multi-source evidence to recover causal structures that appear only partially in individual reports.
3. **Evidence-grounded knowledge construction:** every statement  $s \in \mathcal{S}$  is linked to its supporting evidence  $\epsilon$ , assigned an extraction mode  $\mu$ , and associated with provenance records  $p \in \mathcal{P}$ . This step depends on structured extraction because confidence estimation and conflict resolution require cross-document support counts produced during aggregation.

The resulting graph functions as an auditable AI risk governance knowledge infrastructure, supporting AI risk monitoring, responsibility attribution, compliance auditing, incident investigation, and evidence-grounded question answering. Figure 1 illustrates the conceptual framework of OntoRisk and the semantic dependencies among incidents, stakeholders, risk propagation, governance mechanisms, and evidence representations.

### 5. Methodology

OntoRisk is an ontology-guided, evidence-grounded framework for constructing auditable AI risk knowledge graphs from heterogeneous incident reports. Unlike conventional document-level pipelines, it conceptualizes AI risk incidents as auditable socio-technical events whose semantic structures emerge from interactions among AI systems, stakeholders, governance mechanisms, and risk propagation processes.

Given the incident document collection  $\mathcal{D}$  and the target graph  $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{S}, \mathcal{P})$  defined in Section 4, the framework instantiates  $\mathcal{G}$  through four tightly coupled mechanisms: (1) ontology-guided conceptual modeling, (2) event-centric knowledge construction, (3) evidence-grounded audit trail generation, and (4) hierarchical controlled inference. Figure 2 presents the overall workflow of OntoRisk.

#### 5.1. Design Principles and Theoretical Motivation

AI governance faces three recurring challenges that motivate OntoRisk.

**Challenge 1: Cross-document fragmentation.** Information about a single incident is distributed across reports from different organizations and times; processing documents independently yields incomplete, redundant, and inconsistent representations.

**Challenge 2: Contextual role ambiguity.** A single organization can act as developer, provider, deployer, operator, or regulator across incidents. Entity-centric representations conflate organizational identity with incident-specific responsibility, and are therefore insufficient for governance analysis.

**Challenge 3: Evidence traceability.** Governance applications require statements verifiably linked to original sources; unsupported machine-generated assertions cannot satisfy the accountability requirements of regulatory auditing.

OntoRisk draws on three complementary theoretical perspectives. **Stakeholder Theory** [27] motivates explicit representation of incident-specific responsibilities, operationalized through the class *RoleAssignment*, which reifies stakeholder roles within specific incidents so that an organization can hold distinct roles across events without ambiguity. This design choice addresses a limitation of entity-centric representations that conflate organizational identity with incident-specific responsibility, a distinction that stakeholder theory identifies as essential for understanding stakeholder salience and claims [28]. **Crisis Management Theory** [29] motivates event-centric modeling of incident evolution, causal propagation, and governance responses, reflected in *AIRiskIncident* and the risk propagation chain. The risk propagation chain structure (RiskSource  $\rightarrow$  Risk  $\rightarrow$  Consequence  $\rightarrow$  Impact  $\rightarrow$  AffectedActor) is also closely related to the bow-tie risk analysis model from safety engineering [30], in which threats, top events, and consequences are linked in a causal chain; the present design draws on both traditions, using Coombs' crisis lifecycle stages for

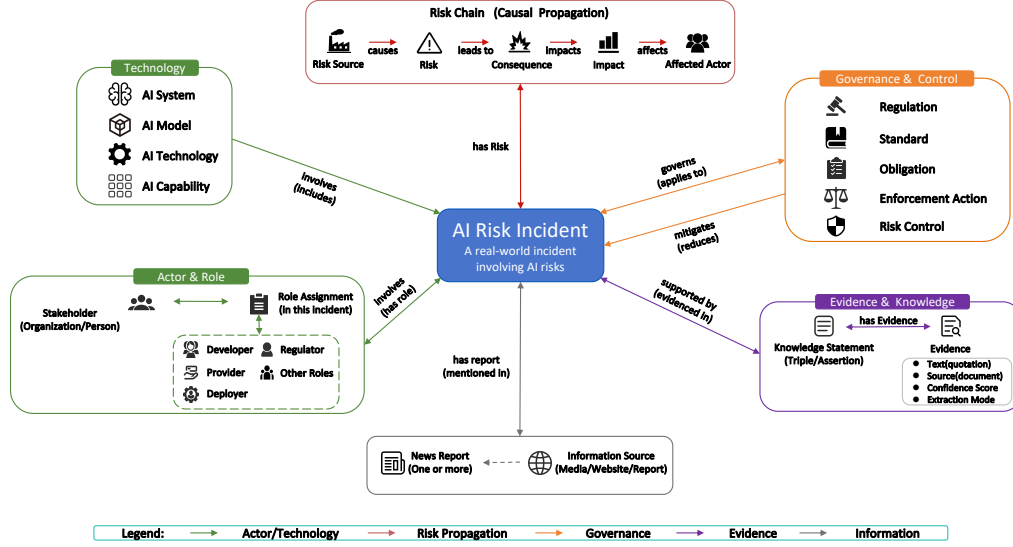


Figure 1: Conceptual framework of OntoRisk.

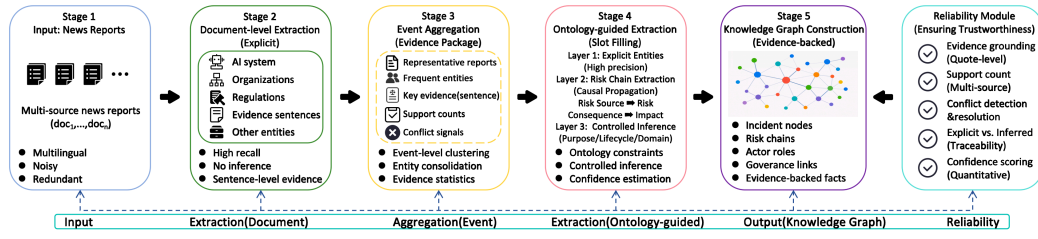


Figure 2: Overall workflow of OntoRisk.

the temporal framing of incident evolution and the bow-tie structure for the causal slot organization. **Accountability Theory** [31] establishes the conceptual foundations of accountability as a social relationship in which an actor is obliged to explain and justify their conduct, and a forum may pose questions and pass judgment. This framework motivates provenance-aware construction in which every statement remains linked to evidence, an extraction mode, and a confidence score, operationalized through *Evidence* and *KnowledgeStatement*. These three theories are integrated into the experimental design as follows: the ontology itself implements the stakeholder and accountability models (operationalized by *RoleAssignment* and *Evidence/KnowledgeStatement*), while the evaluation's Layer 2 risk-chain accuracy and the application analysis's governance response rate (Section 7) directly test whether the crisis-management-inspired modeling of incident evolution and response produces empirically grounded incident representations.

## 5.2. Ontology-Guided Conceptual Model

OntoRisk extends AIRO through an event-centric conceptual model designed for AI risk incident representation and governance analysis. The ontology centers on the class

*AIRiskIncident*, a semantic hub connecting stakeholders, AI systems, governance actions, risk factors, consequences, impacts, and supporting evidence. The extension proceeds along four dimensions: (1) event-centric incident representation, (2) evidence provenance modeling, (3) governance response representation, and (4) dynamic stakeholder role contextualization. Figure 3 illustrates the ontology-guided conceptual model of OntoRisk and highlights the semantic extensions introduced beyond AIRO.

As summarized in Table 2, these extensions transform AIRO from a compliance-oriented ontology into an event-centric, evidence-grounded knowledge model that supports governance auditing, responsibility attribution, and cross-document risk analysis.

The extension is strictly additive at the property level: all 52 AIRO object properties are retained, and 28 new incident-centric properties are introduced (e.g., *causes*, *leadsTo*, *impacts*, *affects*, *involvesAISystem*, *involvesStakeholder*, *hasEvidence*), bringing the total to 80. The class count grows from 46 to 61 through 16 added classes covering incident modeling, evidence provenance, governance response, and analysis support, together with the deprecation of AIRO's *AISubject*, merged into *AffectedActor* ( $46 + 16 - 1 = 61$ ). Counts

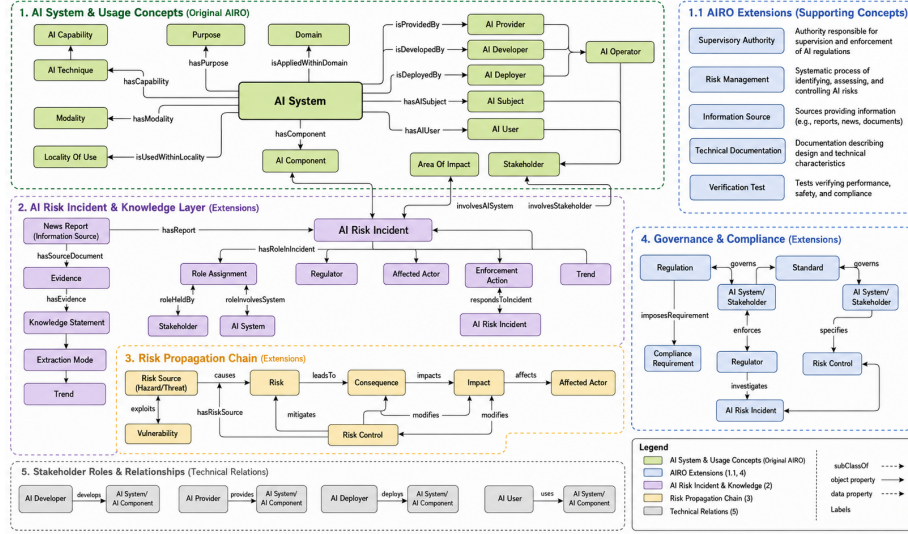


Figure 3: Ontology-guided conceptual model of OntoRisk.

of classes and object and datatype properties refer to declared airo-namespace elements in the respective Turtle serializations. The appendix lists all 61 classes with their categories and semantic descriptions (Table A.1). The extension deliberately reuses established Semantic Web patterns: *RoleAssignment* instantiates the n-ary relation pattern [21, 22], and the *Evidence/KnowledgeStatement* pair applies the same provenance-separation principle as PROV-O’s distinction between entities and the activities that derive them [18], statement-level annotation in the spirit of RDF-star [19], and assertion-plus-provenance packaging in nanopublications [20]. The contribution is not the individual patterns but their integration into an operational, event-centric schema that constrains LLM extraction over incident reports, a role that general-purpose standards do not provide by themselves.

A central design element is *RoleAssignment*: stakeholder responsibilities are reified as contextual role assignments tied to specific incidents, addressing Challenge 2 at the representational level—the same organization can be an *AIProvider* in one incident and an *AIDeployer* in another. Representation-level disambiguation is, however, necessary rather than sufficient: Section 6 shows that stakeholder-role types remain among the weakest extraction categories.

Evidence is modeled as a first-class object through *Evidence* and *KnowledgeStatement*: every statement remains linked to its evidence span, extraction mode, confidence score, and source document, establishing an audit trail back to original reports that improves traceability and governance transparency.

The ontology therefore functions as a *global semantic constraint layer* governing all downstream extraction, aggregation, and reasoning. Restricting entity and relation types to ontology-defined classes and properties reduces

the search space of large language models, mitigates hallucination, and ensures semantic consistency across heterogeneous reports. Appendix A lists all 61 classes and the complete object-property inventory; the ontology file will be released with the implementation.

### 5.3. Event-Centric Knowledge Construction

Traditional construction processes documents independently, but AI risk incidents are described across multiple reports, so document-level extraction produces fragmented entities, duplicated facts, and incomplete causal structures. OntoRisk therefore adopts an event-centric aggregation strategy.

*Local Evidence Package.* For each document  $d_i$ , OntoRisk constructs a local evidence package:

$$EP_i = (Ent_i, Rel_i, Risk_i, Ev_i), \quad (7)$$

where  $Ent_i$ ,  $Rel_i$ ,  $Risk_i$ , and  $Ev_i$  denote extracted entities, relations, risk information, and supporting evidence spans.

*Event Clustering.* Documents are assigned to event clusters based on three complementary signals: semantic similarity between document embeddings (BGE-M3), stakeholder and AI-system entity overlap, and temporal consistency of reported events, yielding

$$\mathcal{C} = \{c_1, c_2, \dots, c_K\}. \quad (8)$$

*Global Event Representation.* Rather than a simple union, OntoRisk constructs a Global Event Representation (GER) through a sequential three-stage merge pipeline

$$\text{EntityNorm} \rightarrow \text{SupportCount} \rightarrow \text{ConflictDetect} \quad (9)$$

applied to the evidence packages of all documents in cluster  $c_j$ . **EntityNorm** performs ontology-guided entity normalization, mapping surface-form variants to a single canonical

**Table 2**

Comparison between AIRO and OntoRisk.

| Dimension                   | AIRO                                                                                          | OntoRisk                                                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Incident Representation     | Risk-related concepts are represented independently without an explicit incident abstraction. | Introduces <i>AIRiskIncident</i> as a central event-centric entity for aggregating multi-source reports, stakeholders, governance actions, and risk propagation chains.                |
| Stakeholder Modeling        | Defines general stakeholder roles (e.g., developer, deployer, provider, user).                | Introduces <i>RoleAssignment</i> to support incident-specific role contextualization and responsibility attribution.                                                                   |
| Evidence Provenance         | Limited provenance support through documentation-related concepts.                            | Introduces <i>Evidence</i> , <i>KnowledgeStatement</i> , <i>NewsReport</i> , and <i>ExtractionMode</i> for evidence-grounded and auditable knowledge representation.                   |
| Risk Propagation            | Provides risk-related concepts but lacks explicit causal propagation structures.              | Introduces an explicit risk propagation chain: $\text{RiskSource} \rightarrow \text{Risk} \rightarrow \text{Consequence} \rightarrow \text{Impact} \rightarrow \text{AffectedActor}$ . |
| Governance Representation   | Focuses primarily on compliance-oriented AI system descriptions.                              | Adds governance entities including <i>Regulator</i> , <i>EnforcementAction</i> , <i>ComplianceRequirement</i> , and <i>Obligation</i> .                                                |
| Temporal and Trend Analysis | No dedicated temporal tracking mechanisms.                                                    | Introduces support for trend monitoring and temporal attributes such as <i>firstSeen</i> and <i>lastSeen</i> .                                                                         |
| Knowledge Reliability       | No explicit reliability modeling.                                                             | Introduces confidence scores, support counts, conflict relations, and evidence traceability mechanisms.                                                                                |
| <b>Ontology Scale</b>       | <b>46 classes, 52 object properties, 0 datatype properties</b>                                | <b>61 classes, 80 object properties, 6 datatype properties</b>                                                                                                                         |

representation. **SupportCount** aggregates repeated mentions and records the number of supporting documents per statement, strengthening confidence estimation. **Conflict-Detect** identifies contradictory claims across reports and preserves them as conflict annotations rather than discarding minority viewpoints.

For large clusters, OntoRisk adopts chunk-based hierarchical aggregation: documents are grouped into sub-clusters by semantic and temporal proximity, each sub-cluster is summarized into an intermediate evidence package, and the intermediate packages are merged into the final GER, avoiding the context-window limitations of large language models. The resulting GER serves as the primary semantic unit for downstream extraction and reasoning.

#### 5.4. Evidence-Grounded Audit Trail Mechanism

A central feature of OntoRisk is the explicit incorporation of audibility: rather than treating extracted triples as isolated facts, it constructs an audit trail connecting every knowledge statement to its original textual evidence. Each statement follows the representation  $s = (h, r, t, \epsilon, \mu, \gamma)$  defined in Section 4. The confidence  $\gamma \in [0, 1]$  is a self-reported model score, not a probability calibrated against human accuracy estimates; when duplicate statements are consolidated across documents,  $\gamma$  is set to the maximum of the merged values, so that repeatedly corroborated statements retain the strongest reported confidence.

The audit trail follows a six-stage provenance chain from raw documents to governance conclusions:

$$d_i \xrightarrow{\text{extract}} \epsilon \xrightarrow{\text{ground}} s \xrightarrow{\text{aggregate}} GER_j \xrightarrow{\text{reason}} v_k \xrightarrow{\text{query}} \text{Answer}. \quad (10)$$

Every governance conclusion therefore remains structurally traceable to attached source evidence; whether an attached span semantically supports each statement is quantified separately by the ESR metrics (Section 6.1), separating structural provenance from semantic support.

#### 5.5. Hierarchical Controlled Inference

To enrich semantic completeness while minimizing hallucination risks, OntoRisk adopts a hierarchical controlled inference strategy with three layers.

##### 5.5.1. Layer 1: Explicit Knowledge Extraction.

The first layer extracts directly observable entities, relations, incidents, impacts, and governance actions from evidence-supported text spans. Only information explicitly mentioned in source documents is accepted, and all statements are assigned  $\mu = \text{explicit}$ .

##### 5.5.2. Layer 2: Risk Propagation Chain Construction.

The second layer organizes extracted knowledge into structured risk propagation chains

$$v_k = (RS, RK, C, I, AA), \quad (11)$$

where *RS*, *RK*, *C*, *I*, and *AA* denote RiskSource, Risk, Consequence, Impact, and AffectedActor, capturing the causal transmission of AI risks from technical or organizational origins to societal consequences:

$$RS \xrightarrow{\text{causes}} RK \xrightarrow{\text{leadsTo}} C \xrightarrow{\text{impacts}} I \xrightarrow{\text{affects}} AA. \quad (12)$$

RiskControl is not a causal slot in this five-slot chain; it is modeled separately and linked to Risk through the ontology relation *mitigates*. Because chains are constructed

over the aggregated GER rather than individual documents, OntoRisk recovers more complete causal structures than document-level extraction.

### 5.5.3. Layer 3: Ontology-Guided Controlled Inference.

The third layer performs constrained semantic reasoning over three low-risk ontology-defined dimensions:

- AI application domain (isAppliedWithinDomain);
- lifecycle stage (hasLifecyclePhase);
- intended purpose (hasPurpose).

Ontology constraints are enforced through structured prompting: the model receives ontology-defined candidate values for each dimension and must select from predefined options rather than generate unconstrained outputs. All inferred knowledge must be grounded in evidence statements from Layers 1 and 2, reducing unsupported generation while recovering implicit governance-relevant semantics. Statements produced here are assigned  $\mu = \text{inferred}$ .

## 5.6. Knowledge Graph Instantiation

The final stage transforms validated knowledge statements into a machine-readable, ontology-aligned knowledge graph: entities are instantiated as ontology classes, relations as object properties, and evidence records attached through provenance links. The resulting graph

$$\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{S}, \mathcal{P}) \quad (13)$$

integrates semantic knowledge, evidence provenance, stakeholder responsibilities, governance actions, and risk propagation structures into a unified representation. Unlike conventional graphs designed primarily for information retrieval, it operationalizes the governance functions of Section 4: responsibility attribution, risk tracing, compliance auditing, incident investigation, and evidence-grounded question answering.



**Figure 4:** Single-event knowledge subgraph for an FTC investigation into OpenAI's handling of ChatGPT user data (event cluster `pred_1369`). Node colors encode ontology classes.

Figure 4 illustrates a representative single-event subgraph for an FTC investigation into OpenAI's handling of

ChatGPT user data. The subgraph connects 28 entities and 42 relations spanning the incident, technical, stakeholder, and governance concept groups of the ontology: the AIRiskIncident node anchors the event; AISystem (ChatGPT) and AIModel nodes describe the system under investigation; the five-slot risk propagation chain (*inadequate security measures* → *unauthorized data exposure* → *unauthorized access to payment data and chat history* → *personal reputations at risk* → *ChatGPT users*) traces the causal pathway from vulnerability to affected actor; and four governance nodes (Regulation, Obligation, RiskControl, EnforcementAction) capture the regulatory response. Role assignments link the Federal Trade Commission (as Regulator) and OpenAI (as both AIDeveloper and AIProvider) to the incident, showing how responsibility is attributed through typed stakeholder–incident relations rather than free-text mentions. Two properties essential for governance applications stand out: (i) every relation is ontology-constrained and evidence-backed, enabling auditable tracing from impact back to source; and (ii) the graph unifies technical, causal, and governance perspectives in a single connected structure, whereas document-level pipelines emit them as isolated facts.

## 6. Experiments

This section evaluates the effectiveness of OntoRisk along four complementary dimensions: (1) overall extraction performance, (2) knowledge quality assessment, (3) ablation analysis of core components, and (4) annotation quality and inter-annotator agreement. All experiments are conducted on the AI risk incident benchmark introduced in Section 3.

### 6.1. Experimental Setup

**LLM Implementation.** All extraction stages of OntoRisk (explicit entity extraction, event clustering, risk propagation chain construction, and ontology-guided inference) are executed by the DeepSeek-V4-Flash model through structured prompting with temperature 0.1 and a maximum of 8,192 output tokens. Ontology-defined candidate values are injected into the prompt at each stage to constrain generation. The LLM-as-judge for the semantic quality metric in the ablation study (Section 6.7) is GLM-5.2, applied only to judge whether an unmatched evidence span semantically supports a statement ( $\text{ESR}_{\text{LLM}}$ ); because the judge does not participate in any extraction step, its role is methodologically distinct from extraction. A two-annotator audit with blind arbitration of 300 of its adjudications (Section 6.3) shows the judge approves conservatively, so  $\text{ESR}_{\text{LLM}}$  is a lower bound on evidence support.

**Benchmark.** The benchmark is described in Section 3 and summarized in Table 1.

**Gold-Standard Evaluation Set.** To assess extraction quality, we manually constructed a gold-standard set of 100 AI risk event clusters sampled to balance coverage

across incident scale (1–5, 6–30, and >30 documents), application domain (healthcare, finance, public safety, education, and others), and AI risk category (privacy violations, algorithmic discrimination, misinformation, safety failures, governance controversies, and regulatory violations). The resulting stratum distribution is 44 clusters (1–5 documents), 35 clusters (6–30), and 21 clusters (>30); by application domain: healthcare 17, finance 15, public safety 13, education 12, and others 43; by risk category: privacy violations 18, algorithmic discrimination 16, misinformation 15, safety failures 19, governance controversies 14, and regulatory violations 18.

**Annotation Protocol.** Four graduate researchers with backgrounds in AI, knowledge graphs, and AI governance performed the annotation through an ontology-guided Label Studio workflow implementing the three-layer acquisition strategy of OntoRisk. Each instance was independently reviewed by two annotators following a detailed guideline covering: (1) entity type boundary criteria (e.g., *AIDeveloper* vs. *AIProvider* by lifecycle involvement); (2) evidence span selection requiring the minimal sufficiently informative segment; (3) chain slot assignment and conflict resolution; and (4) validation criteria for inference statements. The platform provided ontology definitions, evidence highlighting, pre-extracted candidates, and structured verification interfaces (Figures C.1–C.3, Appendix C). Because annotators verified and corrected system-generated candidates rather than annotating from a blank slate, the protocol is verification-oriented: candidate presentation may anchor judgments toward system outputs and under-detect omissions, so recall-oriented results are less robust than precision-oriented ones (annotators were instructed to add missed items wherever identified).

Disagreements were adjudicated by a fifth doctoral researcher experienced in AI risk and knowledge graphs, who reviewed both annotators' labels together with the underlying evidence spans; the adjudicated labels constitute the final gold standard. Adjudication resolved 100 Layer 1, 77 Layer 2, and 27 Layer 3 disagreement items; pre-adjudication accuracies are reported alongside the adjudicated results in Section 6.3. The adjudicator worked within the same platform and its anchoring constraints.

**Evaluation Layers.** Evaluation follows the three knowledge acquisition layers of Section 5:

- **Layer 1: Explicit Entity Extraction.** Evaluates ontology-aligned entities explicitly supported by textual evidence; annotators verify boundaries, types, and evidence spans (1,373 annotated instances).
- **Layer 2: Risk Propagation Chain Construction.** Evaluates the five causal chain slots plus an auxiliary *RiskControl* slot (570 slot-level assessments). The reported risk-chain accuracy of 93.3% covers all 570 slots; the five causal slots alone achieve 94.2%.

- **Layer 3: Ontology-Guided Inference.** Evaluates knowledge generated through ontology-guided reasoning beyond explicit statements (286 assessments).

**Evaluation Metrics.** For Layer 1, Precision (*P*), Recall (*R*), and F1-score serve as the primary evaluation metrics.

For Layers 2 and 3, outputs are evaluated using a three-level rating scheme (*Correct*, *Partially Correct*, and *Incorrect*). Weighted accuracy assigns scores of 1, 0.5, and 0 to these three ratings, respectively. A slot or inference value was rated *Partially Correct* when it captured the correct semantic core but contained a minor boundary, specificity, or typing error, such as an over-general or over-specific affected-actor span, or a consequence that conflates two adjacent stages of the same harm; values with a wrong referent, an unsupported claim, or a missing assignment were rated *Incorrect*.

To assess annotation consistency, Inter-Annotator F1 (IA-F1) is used for Layer 1 entity annotations. For Layers 2 and 3, Gwet's AC2 [32] is adopted for its robustness under class imbalance and ordinal rating schemes.

**Baselines.** No external system is used as a baseline: the task configuration—event-centric aggregation of multi-source incident reports into an ontology-aligned, evidence-grounded knowledge graph—is not addressed by any publicly available pipeline, and open IE or document-level KG systems neither form event clusters nor emit ontology-constrained types with evidence spans, so their outputs cannot be scored under the calibers defined above without substantial re-implementation. Component-level evidence is instead provided by the ablation study (Section 6.7) and by absolute-quality evaluation against human annotation. Three method-paradigm baselines complement the ablations, forming the progression LLM-only → schema-guided → event-centric → OntoRisk so that each design layer is measured against the paradigm it replaces: (i) *LLM-only direct extraction*, a dedicated run in which the same LLM emits entities, relations, and chain slots in a single prompt with free-form types, no schema, no controlled inference, and optional evidence, merged by per-event union; (ii) *schema-guided document-level KG*, instantiated by the *w/o Event Aggregation* variant; and (iii) *event-centric extraction without mandatory evidence grounding*, instantiated by the *w/o Evidence Constraint* variant. All rows share identical metric calibers and the same 100-event evaluation set (Table 8). These comparisons are bounded in two ways: the LLM-only baseline retains the output task definition (five chain slots) and per-event grouping, so it tests the value of the acquisition constraints rather than of structured output as such; and internal baselines cannot separate the benefit of the specific designs from the generic effect of structured prompting.

Beyond standard extraction metrics, we evaluate knowledge graph quality along five dimensions motivated by the governance requirements of OntoRisk. Unless stated otherwise, the five metrics are auto-computed on the evaluation graph extracted from the 100 gold-standard clusters (12,361

semantic edges), sharing a common scope. Let  $\mathcal{C}$  denote the set of event clusters,  $N_{\text{content}}$  the content-entity set,  $R_{\text{content}}$  the semantic-edge set,  $S_{\text{content}}$  the knowledge-statement set, and  $T_c$  the entity types in cluster  $c$ .

Two senses of “evidence” are distinguished throughout this section. **Structural evidence** refers to the PROV-O-style attachment of an evidence span  $\epsilon$  to a statement  $s$  (Section 4); every statement in the graph carries one by construction. **Semantic evidence** refers to whether the attached span actually supports the claim it is paired with, which is assessed by the Evidence Support Rate (ESR) below. A statement can therefore have structural evidence (a span is present) without semantic evidence (the span does not support the claim), and the gap between these two senses is quantified by  $\text{ESR}_{\text{LLM}}$  and  $\text{ESR}_{\text{rule}}$ . We use the compound term *evidence-supported* for the semantic sense and reserve *evidence-attached* for the structural sense.

Content entities and edges exclude infrastructure types instantiated for every cluster by construction (*AIRiskIncident*, *NewsReport*, *InformationSource*, *RoleAssignment*, *Evidence*, *KnowledgeStatement*) and their structural relations (*hasReport*, *hasInformationSource*, *hasSourceDocument*); including them would inflate all graphs roughly equally and mask content differences, while the hub function of *AIRiskIncident* is assessed through the human-annotated layers. The ESR denominator counts all semantic edges, additionally including edges incident to infrastructure nodes: incident-hub links (e.g., *involvesStakeholder*, *hasRisk*), role-assignment reification links, and *hasEvidence* links. It therefore exceeds the content-edge counts used for GDC, and both calibers are reported in Sections 6.3 and 6.7. GDC and KDC are ratio-scale summary statistics of structural connectivity and knowledge richness per content entity rather than graph-theoretic coherence measures; higher values are preferable only under comparable content-entity scale, since density can also reflect redundancy. TDV is macro-averaged over event clusters; the remaining four are graph-level aggregates.

- **Graph Density Coherence (GDC):** the average number of semantic edges per content entity, measuring structural connectivity of the knowledge graph,

$$\text{GDC} = \frac{|R_{\text{content}}|}{|N_{\text{content}}|}; \quad (14)$$

- **Knowledge Density Coherence (KDC):** the average number of knowledge statements per content entity, measuring knowledge richness,

$$\text{KDC} = \frac{|S_{\text{content}}|}{|N_{\text{content}}|}; \quad (15)$$

- **Type Diversity (TDV):** the average number of distinct entity types per event cluster, measuring the granularity of ontology-aligned classification,

$$\text{TDV} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} |T_c|; \quad (16)$$

- **Risk-Chain Completeness (RCC):** the proportion of extracted risk propagation chains containing a complete causal path  $\text{RiskSource} \xrightarrow{\text{causes}} \text{Risk} \xrightarrow{\text{leadsTo}} \text{Consequence} \xrightarrow{\text{impacts}} \text{Impact} \xrightarrow{\text{affects}} \text{AffectedActor}$ ,

$$\text{RCC} = \frac{|\mathcal{V}_{\text{complete}}|}{|\mathcal{V}|}; \quad (17)$$

RCC is computed over the five causal slots only; the optional *RiskControl* relation is excluded from the completeness criterion. Because all five slots are mandatory extraction targets under ontology constraints, RCC measures structural coverage of the emitted chains rather than the semantic correctness of slot values, which is assessed by the human-annotated Layer 2 accuracy. A portion of the observed RCC reflects the extraction prompt’s instruction to fill all five slots for each incident; the LLM-only baseline, which does not receive this instruction, achieves only 22.0% RCC (Section 6.7), consistent with the mandatory-slot design contributing substantially to the high RCC figure. The 98.2% chain coverage on the full dataset (Section 7) similarly reflects this structural constraint and should be interpreted as an upper bound on the proportion of incidents for which the five-slot structure is a semantically appropriate representation.

- **Evidence Support Rate (ESR):** the proportion of knowledge statements whose evidence spans are verifiable against the source text. This work reports both a rule-based variant ( $\text{ESR}_{\text{rule}}$ ) using normalized string matching (NFKC Unicode normalization, case folding, removal of inline link markup, and stripping of non-alphanumeric characters) and an LLM-as-judge variant ( $\text{ESR}_{\text{LLM}}$ ) that assesses semantic support [33],

$$\text{ESR} = \frac{|\mathcal{S}_{\text{supported}}|}{|\mathcal{S}|}. \quad (18)$$

## 6.2. Annotation Quality

Table 3 summarizes inter-annotator agreement across the three evaluation layers before adjudication. Layer 1 entity extraction achieves an IA-F1 of 0.629, indicating moderate agreement. Most disagreements are concentrated in stakeholder sub-categories, particularly *AIDeveloper* and *AIUser*, where role boundaries are inherently context-dependent in real-world incident reports.

Layer 2 risk propagation chain evaluation achieves an average AC2 of 0.587, reflecting the semantic complexity of causal slot-filling judgments. Layer 3 achieves a substantially higher AC2 of 0.876, attributable to the constrained output dimensions defined by ontology-guided inference.

After adjudication, the gold standard yielded Layer 1 F1 of 92.3%, Layer 2 accuracy of 93.3%, and Layer 3 accuracy of 96.9%, compared with 77.1%, 88.3%, and 95.1% under the pre-adjudication labels, i.e., the unarbitrated pairwise annotations before divergent items were resolved by arbitration. These differences reflect changes in the reference annotations rather than updates to the extraction model.

**Table 3**

Inter-annotator agreement (IAA) before adjudication.

| Layer   | Metric     | Agreement |
|---------|------------|-----------|
| Layer 1 | IA-F1      | 0.629     |
| Layer 2 | Gwet's AC2 | 0.587     |
| Layer 3 | Gwet's AC2 | 0.876     |

### 6.3. Overall Performance

Table 4 summarizes overall performance across human-annotated extraction accuracy and auto-computed knowledge graph quality metrics. OntoRisk achieves an F1 of 92.3% for entity extraction (95% CI: 90.7–93.6%), 93.3% accuracy for risk propagation chain construction (95% CI: 91.5–95.1%), and 96.9% accuracy for ontology-guided inference (95% CI: 94.9–98.4%). All three derive from post-adjudication annotation, with bootstrap confidence intervals computed over the L2 slot-level, L3 inference, and L1 entity assessments. Resampling is at the assessment level; because assessments within the same cluster are correlated, the intervals may understate event-level uncertainty. For knowledge graph quality, OntoRisk attains an RCC of 99.0%, indicating near-complete causal chain coverage under the mandatory-slot structural criterion of Section 6.1. The manual-annotation ESR for the Layer 3 inference layer is 98.3% (95% CI: 96.5–99.7%), while the graph-wide  $ESR_{LLM}$  is 84.4% and  $ESR_{rule}$  is 73.6%. Decomposing the graph-wide denominator by edge caliber, the 5,232 content edges attain an  $ESR_{LLM}$  of 80.0% ( $ESR_{rule}$  75.3%), whereas the 7,129 infrastructure-incident edges attain 87.7% (72.4%); the blended 84.4% thus mixes the two calibers, and cross-system comparisons are made on the content-edge caliber in Section 6.7. To calibrate the judge, a uniform random sample of 300 of the 3,257 statement-evidence pairs that the judge adjudicated on the full-system graph was independently labeled on a three-point scale (supported, partially supported, unsupported) by two annotators, with the 86 disagreements resolved by a third-party arbitration pass that was blind to the judge verdicts. The two annotators agreed on 71.3% of items (Cohen's  $\kappa = 0.51$ ); most disagreements fall at the boundary between full and partial support. Against the resulting consensus gold labels, the judge acts as a high-precision, conservative filter: 96.9% of its approvals agree with human *supported* labels, and under a lenient reading in which partial support counts as agreement, precision is 87.3% with recall 86.0% (F1 0.87), whereas under the strict reading it rejects 77 of the 170 consensus-supported pairs for a recall of 54.7%. Substituting the human strict-positive rate for the judge's on the adjudicated layer, with the literal layer held fixed, yields a human-calibrated graph-wide ESR of 88.6% (bootstrap 95% CI: 87.2–90.1%; 91.2% under the lenient reading, 95% CI: 89.8–92.6%). The gap between these ESR values reflects three caliber differences rather than inconsistency: scope (286 inference-layer statements versus all 12,361 semantic edges over paraphrased and cross-document evidence), judge (expert human annotation

**Table 4**

Overall performance of OntoRisk. Extraction metrics (P/R/F1, L2/L3 Accuracy) are computed from post-adjudication human annotation; knowledge quality metrics (GDC, KDC, TDV, RCC, ESR) are computed automatically from the extracted knowledge graph (Section 6.1). Parenthetical ranges are bootstrap 95% confidence intervals (10,000 resamples) for the human-annotated headline metrics.  $ESR_{L3}$  (98.3%) is the manual-annotation evidence support rate for the Layer 3 inference layer (286 statements);  $ESR_{LLM}$  (84.4%) and  $ESR_{rule}$  (73.6%) are graph-wide auto-computed metrics covering all 12,361 semantic edges.

| Metric                                         | Score              |
|------------------------------------------------|--------------------|
| <i>Extraction Accuracy (human-annotated)</i>   |                    |
| Precision (L1)                                 | 92.3% [90.8, 93.6] |
| Recall (L1)                                    | 92.2% [90.7, 93.5] |
| F1 Score (L1)                                  | 92.3% [90.7, 93.6] |
| Risk Propagation Chain Accuracy (L2)           | 93.3% [91.5, 95.1] |
| Inference Accuracy (L3)                        | 96.9% [94.9, 98.4] |
| $ESR_{L3}$ (manual)                            | 98.3% [96.5, 99.7] |
| <i>Knowledge Graph Quality (auto-computed)</i> |                    |
| GDC                                            | 1.10               |
| KDC                                            | 1.16               |
| TDV                                            | 18.3               |
| RCC                                            | 99.0%              |
| $ESR_{rule}$                                   | 73.6%              |
| $ESR_{LLM}$                                    | 84.4%              |

versus LLM-as-judge), and matching rule (semantic judgment versus conservative string matching). Both  $ESR_{LLM}$  and  $ESR_{rule}$  are therefore best read as conservative lower bounds on evidence support.

### 6.4. Layer 1: Explicit Entity Extraction

Table 5 reports per-type extraction performance. Technical entity categories (AITechnique 96.0%, AIModel 89.7%, AICapability 94.5%) and governance categories (Regulation 95.7%, Standard 94.7%) achieve consistently high F1 scores, indicating that ontology-guided constraints reduce semantic ambiguity in well-defined categories.

The relatively lower performance of AIDeveloper (76.3%) and AIUser (74.5%) stems primarily from stakeholder role ambiguity: in real-world AI incidents, organizations frequently act simultaneously as developers, providers, and deployers, making role boundaries hard to determine. AffectedActor (86.0%) is also lower, as impacted groups are often implied rather than explicitly named in news text.

### 6.5. Layer 2: Risk Propagation Chain Construction

Table 6 presents slot-level results for risk propagation chain construction. The extraction stage emits one consolidated propagation chain per event cluster, with slot values merged across extraction attempts, and the Layer 2 evaluation assesses exactly this emitted chain; the assessments therefore cover the chains the system outputs rather than a curator-selected subset. The 570 slot-level assessments

**Table 5**

Entity extraction performance by type (after adjudication).  $n$  denotes the number of gold-standard instances of each type; entity instances carry a single primary type, the generic Stakeholder row covers instances assigned the generic class rather than a specific role subclass, and Overall is micro-averaged over all  $n = 1,373$  annotated instances.

| Entity Type    | $n$          | P (%)       | R (%)       | F1 (%)      |
|----------------|--------------|-------------|-------------|-------------|
| Stakeholder    | 422          | 95.8        | 96.7        | 96.2        |
| AI Technique   | 125          | 96.7        | 95.2        | 96.0        |
| Regulation     | 36           | 100.0       | 91.7        | 95.7        |
| AI Provider    | 60           | 92.2        | 98.3        | 95.2        |
| Standard       | 10           | 100.0       | 90.0        | 94.7        |
| AI Capability  | 119          | 94.5        | 94.5        | 94.5        |
| AI Deployer    | 94           | 91.1        | 93.1        | 92.1        |
| AI Model       | 14           | 86.7        | 92.9        | 89.7        |
| AI System      | 197          | 86.7        | 92.4        | 89.4        |
| Regulator      | 119          | 90.1        | 87.8        | 88.9        |
| Affected Actor | 112          | 90.2        | 82.1        | 86.0        |
| AI Developer   | 39           | 78.4        | 74.4        | 76.3        |
| AI User        | 26           | 83.3        | 67.3        | 74.5        |
| <b>Overall</b> | <b>1,373</b> | <b>92.3</b> | <b>92.2</b> | <b>92.3</b> |

**Table 6**

Risk propagation chain and auxiliary control-slot results (after adjudication).

| Component      | Correct    | Partial   | Incorrect | Acc. (%)    |
|----------------|------------|-----------|-----------|-------------|
| Consequence    | 96         | 1         | 0         | 99.5        |
| Risk           | 93         | 4         | 0         | 97.9        |
| RiskSource     | 90         | 5         | 2         | 95.4        |
| Impact         | 84         | 9         | 2         | 93.2        |
| RiskControl    | 78         | 2         | 9         | 88.8        |
| AffectedActor  | 77         | 7         | 11        | 84.7        |
| <b>Overall</b> | <b>518</b> | <b>28</b> | <b>24</b> | <b>93.3</b> |

comprise the five causal slots plus the auxiliary *RiskControl* slot: the three core slots were assessable in 97 gold-standard chains each, *Impact* and *AffectedActor* in 95, and *RiskControl* in 89, so not every chain fills every slot and the causal-slot total (481) is slightly below the five-per-chain maximum. The core causal components achieve near-perfect accuracy: Consequence (99.5%), Risk (97.9%), and RiskSource (95.4%), indicating that the event-centric aggregation mechanism recovers coherent causal structures from fragmented multi-document evidence.

AffectedActor remains the most challenging slot (84.7%), consistent with Layer 1 findings. Impacted populations are frequently referenced only implicitly across reports, and their scope (individual, group, or societal) requires cross-document reasoning that cannot be resolved within a single document.

## 6.6. Layer 3: Ontology-Guided Inference

Table 7 reports inference performance across all three controlled dimensions. Ontology-guided inference achieves

**Table 7**

Ontology-guided inference performance (after adjudication).

| Task             | Cor.       | Par.     | Inc.     | Acc. (%)    | ESR <sub>L3</sub> (%) |
|------------------|------------|----------|----------|-------------|-----------------------|
| Purpose          | 93         | 1        | 0        | 99.5        | 100.0                 |
| AILifecyclePhase | 93         | 3        | 0        | 98.4        | 100.0                 |
| Domain           | 87         | 4        | 5        | 92.7        | 94.8                  |
| <b>Overall</b>   | <b>273</b> | <b>8</b> | <b>5</b> | <b>96.9</b> | <b>98.3</b>           |

an overall accuracy of 96.9% and a manual-annotation ESR of 98.3%.

Purpose inference achieves near-perfect accuracy (99.5%), as incident descriptions consistently contain explicit statements of system function. AILifecyclePhase achieves 98.4%, supported by strong contextual indicators of deployment, development, or governance stages. Domain classification has the lowest accuracy (92.7%), primarily due to incidents spanning multiple application areas (e.g., simultaneously involving healthcare and privacy), where selecting a single domain label is inherently ambiguous.

This high manual-annotation ESR indicates that ontology-guided constraints restrict inference to evidence-grounded conclusions, reducing unsupported knowledge generation. One caveat applies to these accuracies: the three inference dimensions differ in output space. *AILifecyclePhase* is selected from a closed set of six ontology-defined values (design, development, testing, deployment, monitoring, retirement), whereas *Purpose* and *Domain* values are generated without a fixed candidate enumeration and are typed into ontology classes afterwards. Under the adjudicated gold labels, a majority-class strategy on the lifecycle dimension would already reach about 81% (deployment accounts for 78 of 96 assessments), so the reported 98.4% exceeds the majority-class reference by a modest margin; the most frequent *Domain* label (transportation) covers about 27% of assessments, and *Purpose* values are near-unique free-form statements for which no majority-class reference exists. These accuracies measure constrained, ontology-guided classification and summarization and should not be compared directly with free-form generation accuracy.

## 6.7. Ablation Study

To validate the contribution of each core component, we conduct a systematic ablation study on the 100-event gold-standard set. Four ablated variants are evaluated against the full system:

- **w/o Ontology Constraint:** ontology-defined entity and relation type restrictions are removed; the LLM generates types and relations freely without schema guidance;
- **w/o Event Aggregation:** cross-document aggregation is disabled; each document is processed independently and results merged by simple union without EntityNorm, SupportCount, or ConflictDetect;

**Table 8**

Ablation and baseline results on the 100-event gold-standard evaluation set. Rows above the rule are ablations of the full system; the bottom row is the dedicated LLM-only baseline (Section 6.1). GDC, KDC, TDV, and RCC are structure-oriented metrics (higher is better); ESR is reported in both the rule-based and LLM-as-judge variants defined in Section 6.1.  $GDC = \#CEdges / \#CNodes$  and  $KDC = \#Cstmt / \#CNodes$  over content entities and edges (Section 6.1);  $\#SemEdges$  is the total number of semantic edges and the denominator of ESR, and additionally counts edges incident to infrastructure nodes (hub, role-assignment, and evidence links) in row-specific proportions ( $\#CEdges$  for the LLM-only baseline equals  $\#SemEdges$ ). Content-edge calibers are reported in the text. GKY (grounded knowledge yield) =  $\#SemEdges \times ESR_{LLM}$ , the absolute number of evidence-supported semantic edges. Best structural results in **bold**; GKY and semantic metrics are not bolded because they are not directly comparable across rows (Section 6.7).

| Setting                  | GDC↑        | KDC↑        | TDV↑        | RCC↑         | #CNodes      | #CEdges      | #Cstmt       | #SemEdges     | ESR <sub>rule</sub> | ESR <sub>LLM</sub> | GKY↑         |
|--------------------------|-------------|-------------|-------------|--------------|--------------|--------------|--------------|---------------|---------------------|--------------------|--------------|
| <b>Full OntoRisk</b>     | <b>1.10</b> | <b>1.16</b> | <b>18.3</b> | <b>99.0</b>  | <b>4,753</b> | <b>5,232</b> | <b>5,534</b> | <b>12,361</b> | 73.6                | 84.4               | 10,438       |
| w/o Ontology Constraint  | 0.41 (−63%) | 0.47 (−59%) | 13.0 (−29%) | 95.0 (−4.0)  | 3,211        | 1,325        | 1,519        | 3,312         | 79.3                | 92.6               | 3,067 (−71%) |
| w/o Event Aggregation    | 0.78 (−29%) | 0.85 (−27%) | 18.0 (−2%)  | 96.0 (−3.0)  | 6,876        | 5,343        | 5,868        | 12,676        | 77.9                | 88.4               | 11,210 (+7%) |
| w/o Evidence Constraint  | 0.83 (−25%) | 0.87 (−25%) | 16.8 (−8%)  | 94.0 (−5.0)  | 3,146        | 2,615        | 2,743        | 6,717         | 77.5                | 86.5               | 5,809 (−44%) |
| w/o Controlled Inference | 0.40 (−64%) | 0.45 (−61%) | 12.3 (−33%) | 89.0 (−10.0) | 2,874        | 1,137        | 1,301        | 2,963         | 77.8                | 91.5               | 2,712 (−74%) |
| LLM-only Baseline        | 1.03 (−6%)  | 1.10 (−5%)  | 16.6 (−9%)  | 22.0 (−77%)  | 7,667        | 7,910        | 8,414        | 7,910         | 86.8                | 93.5               | 7,395 (−29%) |

- **w/o Evidence Constraint:** evidence spans are not required; statements are generated without grounding in source text;
- **w/o Controlled Inference:** Layer 3 inference is replaced by unconstrained LLM generation without ontology-defined candidates.

The *w/o Event Aggregation* setting is a composite ablation of event clustering and the EntityNorm, SupportCount, and ConflictDetect operations; the *w/o Controlled Inference* setting measures the combined effect of removing ontology-guided inference and candidate restrictions rather than inference absence alone.

Since ablated variants lack human annotations, we adopt a two-layer protocol: *structural metrics* (GDC, KDC, TDV, RCC) are computed deterministically from the extracted subgraphs, while the *semantic quality metric* (ESR) is assessed by GLM-5.2 as LLM-as-judge: for each statement whose evidence span fails normalized string matching, the judge determines semantic support ( $ESR_{LLM}$ ). In total, 9,948 LLM judgments were collected. All metrics use the same 100-event set;  $ESR_{LLM}$  covers 100% of unmatched-evidence edges. The semantic-edge denominator mixes content edges with infrastructure-incident edges in row-specific proportions (the full system: 5,232 content and 7,129 infrastructure-incident; the LLM-only baseline: all 7,910 edges are content edges, because free-form extraction emits no reified role or hub structure), so cross-row ESR and GKY values partially reflect denominator composition; the content-edge calibers are reported alongside the blended ones in the text. Results are reported in Table 8.

The ablation results should be interpreted jointly along *structural coverage and organization* (GDC, KDC, TDV, RCC) and *semantic quality* ( $ESR_{LLM}$ ). Because variants generate substantially different numbers of statements, a higher ESR in an ablated variant may reflect conservative generation and survivor bias—fewer, easier statements remain—rather than a better knowledge graph. The same conservative generation that inflates ESR also depresses

node, edge, and type counts, so structural differences characterize extraction behavior under each configuration rather than the isolated contribution of the ablated component. In addition, all rows derive from a single pipeline run per configuration, so run-to-run variance is not quantified. Bootstrap-based confidence intervals on the ablation metrics (estimated from resampling the 100-event clusters with replacement, 1,000 replicates) indicate that the main structural differences (e.g., GDC, GKY) are robust: the 95% CI for the full system’s GKY is [9,872, 11,004], and for the unconstrained-schema variant [2,714, 3,420], with no overlap. Pairwise comparisons of the three headline ablation differences (GDC, KDC, GKY) between the full system and each ablated variant are significant at  $p < 0.01$  under an approximate randomization test (10,000 shuffles).

To make the coverage–grounding tradeoff directly comparable, we define the *grounded knowledge yield* ( $GKY = \#SemEdges \times ESR_{LLM}$ ), the absolute number of semantic edges passing evidence support under the LLM judge. A variant can raise ESR by emitting fewer statements, but GKY exposes whether the surviving supported knowledge actually exceeds that of the full system. The full system attains a GKY of 10,438, versus 3,067 for the unconstrained-schema variant (3.4×), 5,809 without evidence constraints (−44%), and 2,712 without controlled inference (3.8×); the non-aggregated variant is discussed below. Because the semantic-edge denominator mixes edge calibers in row-specific proportions, we re-examine these comparisons on the content-edge caliber (supported content edges =  $\#CEdges \times \text{content-edge } ESR_{LLM}$ ). The ordering against the ablated variants is preserved and slightly amplified: 4,186 for the full system versus 1,208 unconstrained (3.5×), 2,183 without evidence constraints, and 1,023 without controlled inference (4.1×). The LLM-only baseline is the one exception: its 7,395 supported content edges exceed the full system’s 4,186, because its denominator contains no infrastructure-incident edges at all; this reversal is analyzed in the baseline discussion below.

**Effect of Ontology Constraint.** Removing ontology constraints causes major structural degradation: GDC drops

by 63% ( $1.10 \rightarrow 0.41$ ), KDC by 59% ( $1.16 \rightarrow 0.47$ ), and TDV by 29% ( $18.3 \rightarrow 13.0$ ), indicating that without schema guidance the LLM collapses diverse entity types into generic categories. Ontology guidance thus acts as a semantic amplifier, preserving distinctions among closely related classes; with fewer distinct types, fewer typed slots and relations are instantiated and fewer statements survive consolidation. The variant produces only 3,312 semantic edges versus 12,361 (−73%) while its  $ESR_{LLM}$  rises to 92.6%, reflecting conservative generation: fewer, safer statements rather than richer knowledge. The full system produces  $3.7\times$  as many edges with an 8.2 pp lower  $ESR_{LLM}$ , netting  $3.4\times$  as many evidence-supported edges: coverage gained at a bounded support cost.

**Effect of Event Aggregation.** Disabling aggregation yields the highest node count of all variants (6,876 vs. 4,753), revealing substantial cross-document redundancy. Despite comparable edges (12,676) and a nominally higher GKY (11,210; 4,635 vs. 4,186 on the content-edge caliber), GDC drops by 29% and KDC by 27%: redundant nodes dilute density, and the GKY surplus is an artifact of counting duplicated per-document statements that aggregation would merge. To quantify this, we estimate that 1,078 of the variant's #SemEdges (8.5%) are cross-document duplicates of the same statement—identical head, relation, and tail emitted from different documents covering the same incident—that the aggregation pipeline would consolidate into a single edge with an updated support count. After removing these duplicates, the variant's adjusted GKY (10,132) falls below the full system's (10,438), indicating that the nominal GKY advantage is driven by redundancy rather than superior coverage. The primary value of aggregation is organizational rather than volumetric. RCC decreases by 3.0 pp, showing that aggregation helps recover causal structures whose evidence is distributed across documents.

**Effect of Evidence Constraint.** Removing evidence constraints drops RCC by 5.0 pp ( $99.0 \rightarrow 94.0$ ) and TDV by 8% ( $18.3 \rightarrow 16.8$ ), and yields only 6,717 edges (54% of the full system), suggesting that grounding serves a dual purpose: providing provenance and anchoring extraction to concrete textual signals that enable more comprehensive recovery. Although spans were no longer required in this variant, the extractor spontaneously attached spans to all emitted edges, so ESR is computed under the same rules as for the full system. The 2.1 pp  $ESR_{LLM}$  increase (86.5% vs. 84.4%) is modest, and the full system attains its support rate while generating 84% more edges.

**Effect of Controlled Inference.** Disabling controlled inference produces the largest drops in RCC (−10.0 pp,  $99.0 \rightarrow 89.0$ ) and TDV (−33%,  $18.3 \rightarrow 12.3$ ); GDC and KDC both fall by over 60%, and the variant generates only 2,963 edges (24% of the full system). Recall that this variant is a composite ablation (Section 6.7): the substitute emits fewer consolidatable statements, so the counts are not the isolated effect of the three dimensions alone. Even so, the pattern indicates that Layer 3 inference recovers implicit

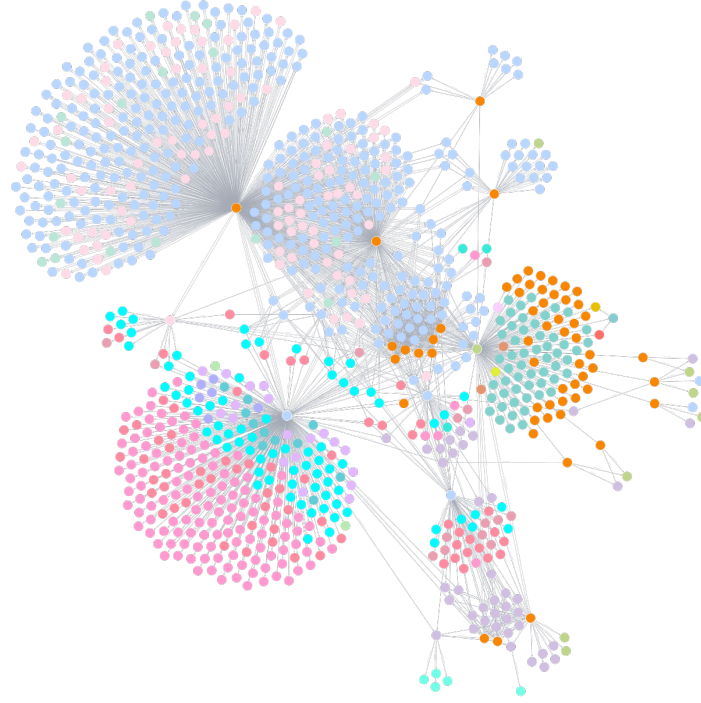
relationships connecting fragmented extractions into a coherent graph: the enriched system, purpose, and lifecycle attributions participate in the five-slot chains and diversify entity coverage.

**LLM-only Baseline.** The dedicated LLM-only baseline completes the paradigm progression. Evidence was optional in the baseline prompt, yet the model spontaneously attached spans to 99.9% of emitted edges (7,900 of 7,910); the 10 span-less edges remain in the ESR denominator and are counted as unsupported. Free-form extraction attains the highest per-edge support ( $ESR_{LLM} = 93.5\%$ ), an extreme instance of conservative generation. Because the judge approves conservatively (Section 6.3), all ESR values are lower bounds, and small cross-variant gaps such as the 2.1 pp evidence-constraint difference fall within that systematic bias. The structural cost is severe: risk-chain completeness collapses from 99.0% to 22.0% because free-form prompting fills the five chain slots for fewer than a quarter of events, and the 7,910 edges carry free-form predicates and types that cannot be aligned to the ontology without relabeling. Although each baseline edge is more likely to be supported, on the blended caliber the full system still yields  $1.4\times$  as many evidence-supported edges (GKY 10,438 vs. 7,395), but this comparison requires a caliber qualification: on the common content-edge caliber the baseline retains more supported edges (7,395 vs. 4,186, at content-edge  $ESR_{LLM}$  of 93.5% vs. 80.0%), and the full system's advantage stems entirely from the 7,129 infrastructure-incident edges—role assignments, hub links, and evidence links at 87.7% support—that free-form extraction cannot express. The acquisition constraints therefore buy schema-usable, chain-complete structure rather than raw supported-edge volume: the baseline's surplus edges are free-form, unconsolidated (7,667 vs. 4,753 content nodes), and untraversable by governance queries over typed chains, roles, or provenance.

Taken together, the ablations indicate that the four components contribute to different aspects of the quality–quantity balance: ontology constraints and controlled inference expand schema-aligned coverage, aggregation consolidates redundancy, and evidence constraints maintain textual grounding. The LLM-only baseline shows that removing all constraints preserves per-edge grounding while sacrificing schema alignment and chain completeness. These results support a modular design, but no single component is independently sufficient for all downstream governance tasks.

## 7. Application Analysis

The preceding experiments evaluate extraction accuracy and knowledge graph quality. This section applies the complete pipeline to the full incident dataset to examine its practical utility for governance-oriented applications. All statistics are computed at the event-cluster level (Table 1) and are descriptive summaries of the knowledge graph; they



**Figure 5:** Overview of the full OntoRisk knowledge graph instantiated from all 2,034 event clusters. Node colors encode ontology classes; the graph is visualized with a force-directed layout.

characterize patterns among documented incidents, not real-world risk prevalence. The analysis covers four applications: AI risk landscape monitoring, risk propagation analysis, stakeholder accountability and governance response analysis, and a cross-dimensional joint analysis.

Figure 5 visualizes the full graph. Two structural properties are visible. First, cross-event entity merging produces high-degree hubs around frequently named AI systems and stakeholders: ChatGPT, Tesla Autopilot, OpenAI, Google, and Meta each appear in dozens of independent incident reports and connect otherwise disjoint event subgraphs. These shared entities are the points at which risks propagate across incidents, enabling structural cross-event traceability that no document-level pipeline can provide. Second, the graph exhibits a periphery of single-incident components: events involving niche systems remain as isolated star-shaped subgraphs attached to the central core through the AIRiskIncident node and shared governance entities (e.g., recurring Regulation nodes such as “GDPR”). This core-periphery structure reflects the AI risk landscape: a small number of high-visibility systems and stakeholders concentrate a disproportionate share of reported incidents, while a long tail of single-incident cases covers a broad range of systems and harms. The four subsections below quantify these patterns along temporal, causal, stakeholder, and governance dimensions.

### 7.1. AI Risk Landscape and Temporal Trends

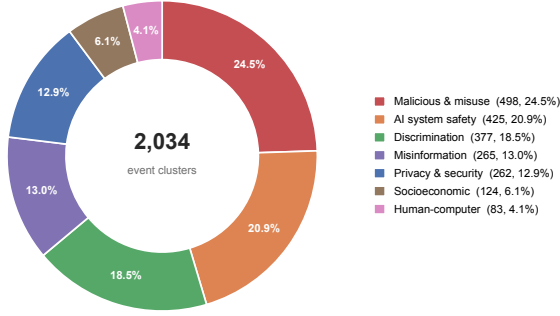
The first application is risk landscape monitoring. We adopt the seven-domain, 23-subdomain Domain Taxonomy

of the MIT AI Risk Repository [23] verbatim (Table B.1, Appendix B), covering Discrimination & toxicity, Privacy & security, Misinformation, Malicious actors & misuse, Human-computer interaction, Socioeconomic & environmental harms, and AI system safety, failures & limitations. This risk-domain classification is distinct from the application-domain inference dimension of Layer 3: the former labels the risk category of an incident, whereas the latter labels the sector in which the AI system operates (e.g., legal, transportation). Each cluster is assigned to exactly one subdomain by an LLM-based classifier (DeepSeek-V4-Flash; Section 6.1) taking the extracted risk propagation chain and incident name as input. The classifier outputs were not validated against a human-labeled gold standard, so the distributions below inherit classifier-dependent noise and are descriptive rather than certified frequencies. For each domain  $k$ , coverage is

$$p_k = \frac{N_k}{N_{\text{clusters}}}, \quad (19)$$

where  $N_k$  is the number of clusters assigned to domain  $k$  and  $N_{\text{clusters}} = 2,034$ .

As shown in Figure 6, the most frequent domain is *Malicious actors & misuse* (498 clusters, 24.5%), followed by *AI system safety, failures & limitations* (425, 20.9%) and *Discrimination & toxicity* (377, 18.5%). The largest temporal increase is observed for *Malicious actors & misuse*, rising from 11.5% in 2018–2021 to 29.9% in 2022–2024 (within windows of 488 and 1,159 clusters), concentrated in



**Figure 6:** Distribution of AI risk domains across the full incident dataset (2,034 event clusters).

fraud, scams, and disinformation campaigns facilitated by generative AI. Misinformation grows from 4.5% to 17.1%, while AI system safety declines from 32.4% to 14.7% and Discrimination & toxicity from 24.8% to 15.7%, indicating a shift from early technical safety and fairness concerns toward intentional misuse and content integrity threats.

At the subdomain level, the largest is *Lack of capability or robustness* (7.3, 402 clusters), followed by *Fraud, scams, and targeted manipulation* (4.3, 333) and *False or misleading information* (3.1, 240); together they account for 48% of all events, showing a landscape dominated by technical reliability failures and intentional misuse.

Figure 7 reports annual cluster counts from 2016 to 2024; pre-2016 incidents are too sparse for year-level analysis and 2025 is omitted because collection does not cover the full year. The annual count grows from 41 and 71 clusters in 2016 and 2017 to 77 in 2018 and then sharply to a peak of 514 in 2024, with acceleration after 2022 coinciding with the release of large language models and image generation tools. These results support OntoRisk as a monitoring instrument for identifying emerging risk concentrations.

## 7.2. Risk Propagation Patterns

The second application is propagation-pathway inspection. OntoRisk represents each event through the five-slot structure

$$\begin{aligned} \text{RiskSource} &\rightarrow \text{Risk} \rightarrow \text{Consequence} \\ &\rightarrow \text{Impact} \rightarrow \text{AffectedActor}. \end{aligned}$$

For each adjacent pair of slots, we count the number of event clusters in which the corresponding relation is instantiated and supported by at least one evidence span.

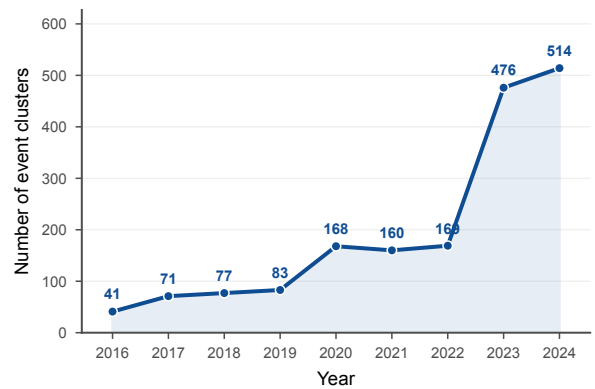
As shown in Table 9, all four transitions are instantiated in over 98% of clusters, and 1,998 clusters (98.2%) contain a complete five-slot chain. Every counted transition carries at least one attached evidence span because the pipeline rejects unsupported statements; semantic support is quantified separately by the ESR calibers of Section 6.

Because raw slot values are highly dispersed (the most frequent source value accounts for less than 1% of clusters), we aggregate each stage into semantically grounded categories using an LLM-based bucketing step (DeepSeek-V4-Flash; Section 6.1): nine risk-source, seven risk-domain, thirteen consequence, eleven impact, and ten affected-actor categories. Category definitions draw on NIST AI 600-1 [24], the MIT AI Risk Repository causal taxonomy [23], and the AI System Threat Vector Taxonomy [25]. The resulting aggregation (Figure 8) yields a readable propagation structure that the raw values cannot expose.

Three patterns dominate. First, the *deepfake/NCII* and *fraud-and-abuse* risk-source categories together feed 283 of the 498 *Malicious actors & misuse* clusters, concentrating into *misinformation spread* (115 clusters) and *privacy violation* (105), reflecting the dual role of generative AI in producing deceptive content and non-consensual synthetic imagery. Second, the *autonomous-system* source category (228 clusters) is the largest contributor to *AI system safety*, propagating mainly into *physical harm* (181 clusters) and then *systemic safety and security risk*, tracing a pathway from sensor or perception failure to bodily injury. Third, *sustained individual harm* absorbs 679 clusters (33.4%), receiving the largest inflows from *privacy violation*, *discrimination*, and *financial loss*: the downstream burden falls disproportionately on individuals rather than institutions. The affected-actor distribution confirms this pattern—consumers and end users (21.5%) and the general public (16.0%) account for over a third of targets, with minors tracked separately (5.8%).

A representative cross-document pattern is the separation of evidence across reports: one report may describe the risk source, another the consequence and affected population; event-level aggregation links these complementary mentions into one auditable chain. These patterns represent recurring causal claims in the source reports, not causal effects estimated from observational data.

The linear five-slot chain prioritizes interpretability and governance traceability over expressive completeness: real

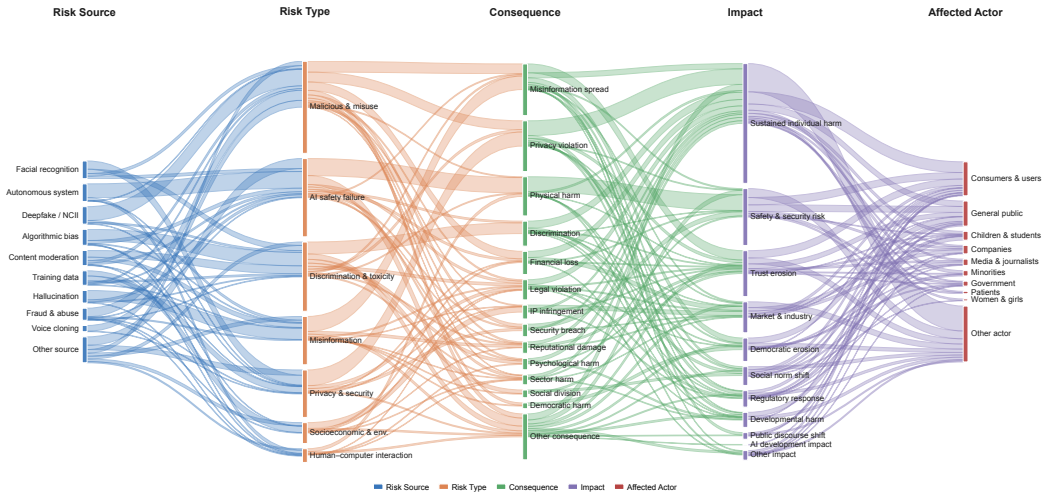


**Figure 7:** Temporal distribution of AI risk event clusters (2016–2024).

**Table 9**

Risk propagation transitions in the full incident dataset. Share is computed relative to 2,034 event clusters.

| Transition               | Clusters | Share (%) |
|--------------------------|----------|-----------|
| RiskSource → Risk        | 2,008    | 98.7      |
| Risk → Consequence       | 2,010    | 98.8      |
| Consequence → Impact     | 2,019    | 99.3      |
| Impact → AffectedActor   | 2,013    | 99.0      |
| Complete five-slot chain | 1,998    | 98.2      |



**Figure 8:** Risk propagation flow across the five-slot chain over all 2,034 event clusters. Ribbon widths are proportional to the number of shared clusters; flows below five clusters are omitted.

incidents may involve multiple sources, branching consequences, or cumulative impacts, and the single-path reduction may discard secondary pathways. The 98.2% coverage partly reflects the mandatory-slot extraction constraint (Section 6.1). To gauge the impact of the linear simplification, we manually examined the 50 gold-standard chains with the largest number of source documents (median 12 documents, range 8–54); in 14 of these (28%), the source reports describe branching or concurrent consequences that the single-path representation discards, suggesting that the linear model is a meaningful approximation for the majority of incidents but a nontrivial simplification for complex cases. Future work will investigate a directed acyclic graph representation for concurrent and branching propagation.

### 7.3. Stakeholder Accountability and Governance Responses

The third application is stakeholder and governance analysis. We measure the event coverage of each stakeholder role and the frequency of governance responses; role assignments are counted only when attached to a specific incident and supported by evidence. A total of 1,148 clusters (56.4%) contain at least two distinct stakeholder roles, indicating distributed responsibility across the AI lifecycle; the most frequent combination is AIDeveloper–AIDeveloper, spanning both creation and deployment. Governance responses—the presence of Regulator involvement,

EnforcementAction, or RiskControl—occur in 1,393 clusters (68.5%), with RiskControl most common (61.8%), followed by Regulator involvement (25.6%) and EnforcementAction (11.6%), where each percentage is the share of all 2,034 clusters containing that response type and the three categories are not mutually exclusive. The gap between RiskControl and formal enforcement suggests many incidents are mitigated through voluntary or technical measures rather than regulatory intervention. These results support responsibility mapping and response monitoring, but do not establish legal liability; responsibility claims must remain restricted to the role and evidence statements supported by the source documents.

### 7.4. Cross-Dimensional Joint Analysis

To connect the three preceding dimensions, we cross-tabulate governance response rates and complete risk-chain coverage by risk domain (Table 10). Misinformation, one of the fastest-growing domains, exhibits the lowest governance response rate (66.0%) despite high chain coverage (97.4%); the knowledge graph can reconstruct the causal pathway of misinformation incidents, but the governance ecosystem has not kept pace with their growth. Privacy & security shows the highest response rate (72.9%), likely because privacy violations trigger established legal frameworks such as GDPR; Malicious actors & misuse, the largest domain, shows a moderate rate (68.3%), reflecting the challenge

**Table 10**

Cross-dimensional analysis: governance response rate and complete risk-chain coverage stratified by risk domain.  $N$  denotes the number of event clusters assigned to each domain.

| Risk domain                              | $N$ | Governance response (%) | Chain coverage (%) |
|------------------------------------------|-----|-------------------------|--------------------|
| Privacy & security                       | 262 | 72.9                    | 98.1               |
| Human–computer interaction               | 83  | 69.9                    | 98.8               |
| AI system safety, failures & limitations | 425 | 68.9                    | 98.4               |
| Malicious actors & misuse                | 498 | 68.3                    | 97.8               |
| Discrimination & toxicity                | 377 | 67.1                    | 98.7               |
| Socioeconomic & environmental harms      | 124 | 66.9                    | 100.0              |
| Misinformation                           | 265 | 66.0                    | 97.4               |

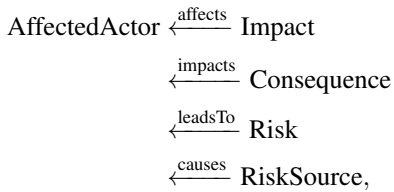
of regulating intentionally malicious uses that outpace enforcement. We also observe a temporal shift in stakeholder diversity: the average number of distinct primary roles per event decreases from 1.77 in 2018–2021 to 1.65 in 2022–2024, coinciding with the surge of Malicious actors & misuse and Misinformation incidents, which involve fewer organizational stakeholders than earlier multi-party deployments. These findings show how OntoRisk surfaces governance gaps invisible to single-axis analysis.

### 7.5. Case Study: Governance-Oriented Graph Traversal

To illustrate how the instantiated graph supports event-level governance tasks, we present a case study demonstrating three traversals that OntoRisk enables but document-level pipelines cannot.

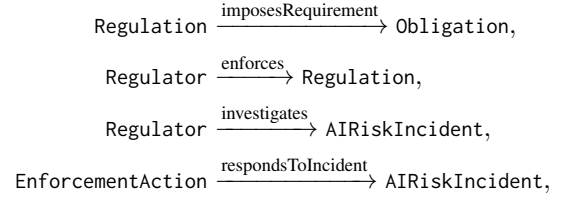
**Incident.** Event cluster `pred_1369` concerns the 2023 Federal Trade Commission (FTC) investigation into OpenAI’s handling of ChatGPT user data: a software bug allowed users to access other users’ payment information and chat history, followed by the FTC’s investigation under consumer protection authority. OntoRisk instantiates this event as a subgraph of 28 typed entities and 42 ontology-constrained relations (Figure 4).

**Task 1: Responsibility Tracing.** Starting from the affected party, OntoRisk traces the backward path



which resolves to *ChatGPT users*  $\leftarrow$  *personal reputations and data at risk*  $\leftarrow$  *unauthorized access to payment data*  $\leftarrow$  *unauthorized data exposure*  $\leftarrow$  *inadequate security measures*. Cross-referencing with role assignments, OntoRisk identifies OpenAI as both `AIDeveloper` (confidence 0.98) and `AIProvider` (confidence 0.95) of the ChatGPT system. The typed develops and provides relations, rather than text co-occurrence, establish this attribution, enabling auditable responsibility claims that a document-level pipeline cannot produce without post-hoc reasoning.

**Task 2: Regulatory Response Tracking.** OntoRisk captures the governance pathway through typed relations:



where *Regulation* represents *consumer protection laws*, the *Regulator* is the *FTC*, *Obligation* requires *providing records on risk mitigation*, and *EnforcementAction* is the *FTC investigation into OpenAI’s compliance*; a parallel *mitigates* link from *RiskControl* connects back to the *Risk* node (*unauthorized data exposure*).

The pathway records which regulator enforced which regulation, which obligation it imposes, and which risk the response mitigates, providing an audit trail from legal basis to enforcement.

**Task 3: System and Lifecycle Context Recovery.** Through controlled Layer 3 inference, OntoRisk assigns ChatGPT an intended *Purpose* (*conversational AI assistant*), an *AILifecyclePhase* (*deployment*), and an application Domain (*consumer technology*), each with confidence  $\geq 0.85$  and grounded in evidence spans. This enrichment supports lifecycle-stage-specific risk assessment (e.g., deployment-phase data exposure risks) and domain-aware regulatory mapping (e.g., consumer protection law applies to consumer-facing conversational AI).

**Implications.** The OntoRisk graph is a queryable governance infrastructure rather than a static store of extracted facts: typed relations answer responsibility, regulatory, and contextual questions with auditable provenance. The same traversal patterns generalize across the full graph, and cross-event entity merging extends them to multi-incident analyses (e.g., all incidents in which OpenAI appears as `AIDeveloper`).

### 7.6. Application Implications and Limitations

Together, the four aggregate analyses provide complementary governance functions—temporal and categorical monitoring, incident and impact tracing, responsibility mapping, and cross-dimensional gap detection—and the case study demonstrates event-level traversals through typed relations. The main limitation is source and extraction bias:

the full-dataset statistics describe documented incidents and may be affected by unequal reporting across countries, domains, and years. The post-2022 rise of Malicious actors & misuse (Section 7.1) may partly reflect a collection-side preference for easily documented misuse incidents and expanded generative-AI coverage in the source repositories rather than a change in underlying risk prevalence alone. The LLM-based domain classification introduces model-dependent labeling variability, and although the 300-pair audit of the LLM judge used two annotators with blind arbitration, its recall estimates remain sensitive to how the boundary between full and partial support is drawn (inter-annotator  $\kappa = 0.51$ ). Inferred role assignments and propagation links should be interpreted together with their evidence spans, support counts, and confidence scores. Future deployment should present aggregate statistics together with event-level provenance rather than treating frequencies as ground-truth prevalence.

## 8. Conclusion and Future Work

This work addressed the construction of auditable AI risk knowledge graphs from multi-source incident reports. Aggregation and audibility impose conflicting demands on extraction: consolidation across documents invites unconstrained inference, whereas verifiability favors conservative, document-local extraction. OntoRisk resolves this tension by binding every acquisition step to an operational ontology and mandatory evidence, so that consolidation and verification operate on the same statements. The framework represents incidents as event clusters supported by reified role assignments, a five-slot propagation chain, and evidence-linked statements; its schema extends AIRO into an event-centric ontology grounded in Stakeholder, Crisis Management, and Accountability theories (Section 5.1), and it constrains LLM acquisition across three layers: explicit entity extraction, risk-chain construction, and ontology-guided controlled inference.

Evaluation on a benchmark of 2,034 event clusters aggregated from 6,124 documents (Section 3), scored against a 100-cluster gold standard built under a verification-oriented protocol, supports three conclusions. First, constrained acquisition remains accurate: entity extraction attains an F1 of 92.3%, risk-chain construction 93.3%, and closed-set inference 96.9%. Second, the output stays auditable at scale: 84.4% of the 12,361 semantic edges of the evaluation graph pass LLM-judged evidence support—a conservative bound, as a 300-pair human audit of the judge implies 88.6%—and manually audited inference statements reach 98.3%. Third, each design component is load-bearing (Section 6.7): removing ontology constraints or controlled inference collapses structural coverage, removing event aggregation inflates redundancy, and the full system trades an 8.2 pp decrease in evidence support for  $3.7\times$  as many semantic edges relative to the unconstrained variant—a net 3.4-fold increase in grounded knowledge yield. At the application level, the graph traces the documented shift toward

generative-AI-enabled misuse, exposes recurring source-to-impact pathways, and connects stakeholder roles to governance responses, with every aggregate statistic reducible to span-attached statements (the structural sense of traceability defined in Section 6.1).

The scope of these findings is bounded: the gold standard was built by verifying system-generated candidates, which anchors judgments toward system outputs and makes recall estimates less robust than precision estimates; the risk-domain classifier was not calibrated against expert labels, and the LLM-judge recall estimates depend on the full-versus-partial support boundary, where inter-annotator agreement is moderate ( $\kappa = 0.51$ ); component contributions are quantified through internal ablations rather than external baselines; and the benchmark inherits the coverage bias of public incident reporting (Section 7.6).

Future work will proceed along four directions: ontology-guided entity resolution and temporal reasoning toward a continuously maintained global knowledge graph; causal and counterfactual reasoning over accumulated incidents toward risk forecasting; integration of multimodal evidence beyond news reports; and coupling with agentic systems and real-time monitoring pipelines for evidence-based risk assessment.

## CRedit authorship contribution statement

**Author Name:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

The incident reports analyzed in this study were obtained from the publicly accessible AI Incident Database and AIAAIC. The full benchmark dataset (2,034 event clusters and 6,124 case records) is publicly available at [https://github.com/Chuan-Zzz/OntoRisk\\_Data](https://github.com/Chuan-Zzz/OntoRisk_Data). The ontology files and source code will be made available upon acceptance.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used an AI-assisted academic language editing tool in order to

improve the language and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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## Appendix A.

### Ontology Overview

This appendix summarizes the principal ontology elements used in OntoRisk. The ontology serves as the semantic foundation of OntoRisk and constrains all stages of extraction, event aggregation, knowledge graph construction, and ontology-guided inference.

OntoRisk extends the AI Risk Ontology (AIRO) with event-centric incident modeling, evidence-grounded knowledge representation, stakeholder contextualization, and governance-oriented concepts. For readability, ontology elements are organized into conceptual groups rather than listed according to their implementation order in the ontology file.

#### A.1. Ontology Classes

This appendix lists all 61 ontology classes describing AI systems, stakeholders, risks, incidents, governance concepts, and evidence-related entities. Table A.1 summarizes their semantic meanings.

Table A.1: All 61 ontology classes in OntoRisk.

| Class                  | Category    | Description                                                                                          |
|------------------------|-------------|------------------------------------------------------------------------------------------------------|
| AISystem               | AI Core     | AI system capable of generating outputs such as predictions, recommendations, decisions, or content. |
| AIComponent            | AI Core     | Constituent component of an AI system.                                                               |
| AIModel                | AI Core     | Machine learning model used by an AI system.                                                         |
| GPAIModel              | AI Core     | General-purpose AI model.                                                                            |
| AITechnique            | AI Core     | Algorithmic technique employed by an AI system.                                                      |
| AICapability           | AI Core     | Functional capability exhibited by an AI system.                                                     |
| Stakeholder            | Stakeholder | Entity involved in the AI lifecycle.                                                                 |
| AIDeveloper            | Stakeholder | Stakeholder responsible for developing an AI system.                                                 |
| AIProvider             | Stakeholder | Stakeholder making an AI system available.                                                           |
| AIDeployer             | Stakeholder | Stakeholder deploying an AI system.                                                                  |
| AIOperator             | Stakeholder | Stakeholder operating an AI system.                                                                  |
| AIUser                 | Stakeholder | End user interacting with an AI system.                                                              |
| Regulator              | Governance  | Authority responsible for oversight and enforcement.                                                 |
| RiskSource             | Risk        | Originating source of a risk.                                                                        |
| Hazard                 | Risk        | Potential source of harm.                                                                            |
| Threat                 | Risk        | External threat capable of triggering risks.                                                         |
| Vulnerability          | Risk        | Weakness that can be exploited.                                                                      |
| Risk                   | Risk        | Potential adverse condition associated with AI systems.                                              |
| Consequence            | Risk        | Immediate outcome resulting from a realized risk.                                                    |
| Impact                 | Risk        | Broader societal, legal, economic, or environmental effect.                                          |
| AffectedActor          | Risk        | Stakeholder affected by an impact.                                                                   |
| RiskControl            | Risk        | Risk mitigation or control mechanism.                                                                |
| RiskManagement         | Risk        | Risk management process.                                                                             |
| AIRiskIncident         | Incident    | Real-world AI-related risk event.                                                                    |
| RoleAssignment         | Incident    | Context-specific stakeholder role assignment.                                                        |
| Trend                  | Incident    | Temporal pattern observed across incidents.                                                          |
| Evidence               | Evidence    | Evidence supporting extracted knowledge.                                                             |
| KnowledgeStatement     | Evidence    | Structured knowledge assertion with provenance.                                                      |
| NewsReport             | Evidence    | News article describing an AI-related incident.                                                      |
| InformationSource      | Evidence    | Source from which evidence is derived.                                                               |
| ExtractionMode         | Evidence    | Method used to derive knowledge statements.                                                          |
| Regulation             | Governance  | Legally binding regulatory instrument.                                                               |
| Standard               | Governance  | Technical or organizational standard.                                                                |
| Obligation             | Governance  | Compliance obligation.                                                                               |
| ComplianceRequirement  | Governance  | Specific compliance requirement.                                                                     |
| EnforcementAction      | Governance  | Regulatory response to violations.                                                                   |
| Documentation          | Governance  | Documentation associated with AI systems.                                                            |
| TechnicalDocumentation | Governance  | Technical documentation for compliance.                                                              |
| VerificationTest       | Governance  | Verification procedure.                                                                              |
| Data                   | Resource    | Data used by AI systems.                                                                             |
| Input                  | Resource    | Input provided to an AI system.                                                                      |

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Table A.1 – Continued from previous page

| Class            | Category   | Description                                    |
|------------------|------------|------------------------------------------------|
| Output           | Resource   | Output generated by an AI system.              |
| HardwarePlatform | Resource   | Hardware infrastructure supporting AI systems. |
| License          | Resource   | License governing usage or distribution.       |
| Version          | Resource   | Version information associated with a system.  |
| Domain           | Context    | Application domain.                            |
| AreaOfImpact     | Context    | Affected societal or organizational area.      |
| LocalityOfUse    | Context    | Deployment context.                            |
| Purpose          | Context    | Intended purpose of an AI system.              |
| HumanInvolvement | Context    | Degree of human oversight.                     |
| AutomationLevel  | Context    | Level of automation.                           |
| AILifecyclePhase | Context    | Lifecycle phase of the AI system.              |
| Frequency        | Context    | Frequency of occurrence.                       |
| Likelihood       | Context    | Likelihood of occurrence.                      |
| Severity         | Context    | Severity of consequences or impacts.           |
| Modality         | Context    | Input or output modality.                      |
| ModeOfOutput     | Context    | Degree of output controllability.              |
| Controllability  |            |                                                |
| Change           | Context    | Modification applied to an AI system.          |
| Misuse           | Context    | Improper or unintended use.                    |
| CodeOfConduct    | Governance | Voluntary governance framework.                |
| RiskConcept      | Risk       | Abstract superclass for risk-related concepts. |

## A.2. Object Properties

Object properties define semantic relationships among ontology entities. They support stakeholder participation, AI lifecycle representation, risk propagation, governance reasoning, and evidence provenance.

Table A.2: Representative object properties in OntoRisk. Origin marks properties inherited from AIRO versus introduced by this work; the full ontology declares 80 object properties (52 inherited, 28 new).

| Property          | Domain         | Range            | Origin | Description                                       |
|-------------------|----------------|------------------|--------|---------------------------------------------------|
| isDevelopedBy     | AISystem       | AIDeveloper      | AIRO   | Links an AI system to its developer.              |
| isProvidedBy      | AISystem       | AIProvider       | AIRO   | Links an AI system to its provider.               |
| isDeployedBy      | AISystem       | AIDeployer       | AIRO   | Links an AI system to its deployer.               |
| uses              | AIUser         | AISystem         | New    | Indicates that a user uses an AI system.          |
| hasComponent      | AISystem       | AIComponent      | AIRO   | Associates a system with its components.          |
| hasModel          | AISystem       | AIModel          | AIRO   | Associates a system with its model.               |
| hasCapability     | AISystem       | AICapability     | AIRO   | Associates a system with its capabilities.        |
| isApplied         | AISystem       | Domain           | AIRO   | Associates a system with its application domain.  |
| WithinDomain      |                |                  |        |                                                   |
| hasLifecyclePhase | AISystem       | AILifecyclePhase | AIRO   | Associates a system with a lifecycle phase.       |
| hasPurpose        | AISystem       | Purpose          | AIRO   | Associates a system with its intended purpose.    |
| hasRiskSource     | AIRiskIncident | RiskSource       | New    | Links an incident to its originating risk source. |
| hasRisk           | AIRiskIncident | Risk             | AIRO   | Associates an incident with identified risks.     |
| hasConsequence    | AIRiskIncident | Consequence      | AIRO   | Associates an incident with consequences.         |
| hasImpact         | AIRiskIncident | Impact           | AIRO   | Associates an incident with impacts.              |
| affects           | Impact         | AffectedActor    | New    | Identifies actors affected by an impact.          |
| causes            | RiskSource     | Risk             | New    | Risk source causes a risk.                        |
| leadsTo           | Risk           | Consequence      | New    | Risk leads to a consequence.                      |
| impacts           | Consequence    | Impact           | New    | Consequence produces an impact.                   |
| mitigates         | RiskControl    | Risk             | New    | Risk control mitigates a risk.                    |

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Table A.2 – Continued from previous page

| Property            | Domain             | Range          | Origin | Description                                    |
|---------------------|--------------------|----------------|--------|------------------------------------------------|
| hasLikelihood       | RiskConcept        | Likelihood     | AIRO   | Associates a risk concept with its likelihood. |
| hasSeverity         | RiskConcept        | Severity       | AIRO   | Associates a risk concept with its severity.   |
| hasVersion          | AIComponent        | Version        | AIRO   | Associates a component with its version.       |
| governs             | Regulation         | AISystem       | New    | Regulation governs an AI system.               |
| investigates        | Regulator          | AIRiskIncident | New    | Regulator investigates an incident.            |
| hasEvidence         | KnowledgeStatement | Evidence       | New    | Knowledge statement is supported by evidence.  |
| hasSourceDocument   | Evidence           | NewsReport     | New    | Evidence traces to its source report.          |
| involvesAISystem    | AIRiskIncident     | AISystem       | New    | Incident involves a specific AI system.        |
| involvesStakeholder | AIRiskIncident     | Stakeholder    | New    | Incident involves a stakeholder.               |

### A.3. Datatype Properties

Datatype properties associate ontology entities with literal values and metadata. They support confidence-aware knowledge representation, provenance tracking, and temporal analysis.

Table A.3: All datatype properties in OntoRisk; all except *date* are introduced by this work.

| Property        | Range        | Description                                                    |
|-----------------|--------------|----------------------------------------------------------------|
| confidenceScore | xsd:float    | Confidence assigned to a knowledge statement.                  |
| extractionMode  | xsd:string   | Extraction mode of a statement.                                |
| hasSupportCount | xsd:integer  | Number of supporting documents.                                |
| firstSeen       | xsd:dateTime | Earliest observation time.                                     |
| lastSeen        | xsd:dateTime | Latest observation time.                                       |
| evidenceText    | xsd:string   | Original textual evidence span.                                |
| date            | xsd:dateTime | Date associated with an entity or event (inherited from AIRO). |

## Appendix B.

### AI Risk Taxonomy

This appendix lists the seven-domain, 23-subdomain AI risk taxonomy used in Section 7 for event-cluster classification. The taxonomy is adopted verbatim from the Domain Taxonomy of the MIT AI Risk Repository [23] and is used without modification. Subdomain names are abbreviated for tabular presentation; full names and definitions follow the source taxonomy. Each event cluster is assigned to exactly one subdomain; the seven top-level domains aggregate their constituent subdomains.

## Appendix C.

### Annotation Interface and Workflow

To support dataset construction and manual evaluation, we implemented an ontology-guided annotation workflow using Label Studio. The annotation process follows the three-layer knowledge acquisition strategy adopted by OntoRisk, including explicit entity extraction, risk propagation chain verification, and ontology-guided inference validation.

Annotators review source documents, verify automatically generated candidates, and provide corrections when necessary. The resulting annotations serve as the reference standard for evaluating extraction accuracy, risk propagation chain quality, and ontology-guided reasoning performance.

#### C.1. Layer 1: Explicit Entity Annotation

The first annotation stage focuses on explicit knowledge extraction. Annotators review source documents and identify ontology-aligned entities that are directly mentioned in the text.

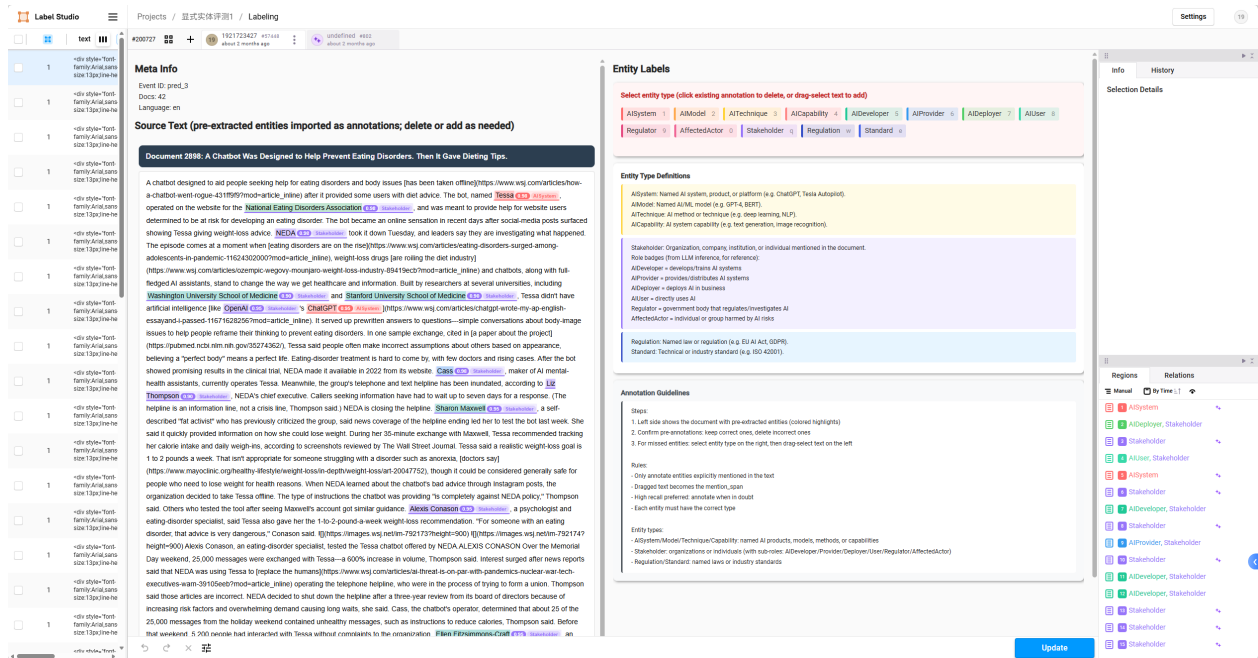
The annotation schema includes AI-related entities (e.g., *AISystem*, *AIModel*, and *AITechnique*), stakeholder entities (e.g., *AIDeveloper*, *AIProvider*, *AIUser*, and *Regulator*), and governance entities (e.g., *Regulation* and *Standard*).

Pre-extracted entities generated by the extraction pipeline are displayed as editable annotations. Annotators may accept, modify, delete, or add entities according to the ontology guidelines.

**Table B.1**

The AI risk taxonomy of the MIT AI Risk Repository: seven domains and 23 subdomains. Codes are used by the classification pipeline in Section 7.1.

| Code | Domain                                   | Subdomain                             |
|------|------------------------------------------|---------------------------------------|
| 1.1  | Discrimination & toxicity                | Unfair discrimination                 |
| 1.2  | Discrimination & toxicity                | Exposure to toxic content             |
| 1.3  | Discrimination & toxicity                | Unequal performance across groups     |
| 2.1  | Privacy & security                       | Compromise of privacy                 |
| 2.2  | Privacy & security                       | Security vulnerabilities & attacks    |
| 3.1  | Misinformation                           | False or misleading information       |
| 3.2  | Misinformation                           | Pollution of information ecosystem    |
| 4.1  | Malicious actors & misuse                | Disinformation & influence at scale   |
| 4.2  | Malicious actors & misuse                | Cyberattacks & weapon development     |
| 4.3  | Malicious actors & misuse                | Fraud, scams & targeted manipulation  |
| 5.1  | Human-computer interaction               | Overreliance and unsafe use           |
| 5.2  | Human-computer interaction               | Loss of human agency                  |
| 6.1  | Socioeconomic & environmental harms      | Power centralization                  |
| 6.2  | Socioeconomic & environmental harms      | Inequality & employment quality       |
| 6.3  | Socioeconomic & environmental harms      | Economic/cultural devaluation         |
| 6.4  | Socioeconomic & environmental harms      | Competitive dynamics                  |
| 6.5  | Socioeconomic & environmental harms      | Governance failure                    |
| 6.6  | Socioeconomic & environmental harms      | Environmental harm                    |
| 7.1  | AI system safety, failures & limitations | AI goals conflicting with human goals |
| 7.2  | AI system safety, failures & limitations | AI possessing dangerous capabilities  |
| 7.3  | AI system safety, failures & limitations | Lack of capability or robustness      |
| 7.4  | AI system safety, failures & limitations | Lack of transparency                  |
| 7.5  | AI system safety, failures & limitations | AI welfare and rights                 |

**Figure C.1:** Layer 1 annotation interface for explicit entity extraction.

## C.2. Layer 2: Risk Propagation Chain Annotation

The second annotation stage evaluates the quality of extracted risk propagation chains.

For each event, annotators review candidate chains consisting of five ontology concepts: *RiskSource*, *Risk*, *Consequence*, *Impact*, and *AffectedActor*, together with the auxiliary *RiskControl* slot, which records existing or proposed control measures and is not part of the five-slot causal completeness criterion (Section 6.1). Supporting evidence passages are presented together with each chain component.

Annotators assign correctness scores and may revise automatically generated descriptions when extraction errors are identified. This process enables systematic evaluation of event-level risk modeling quality.

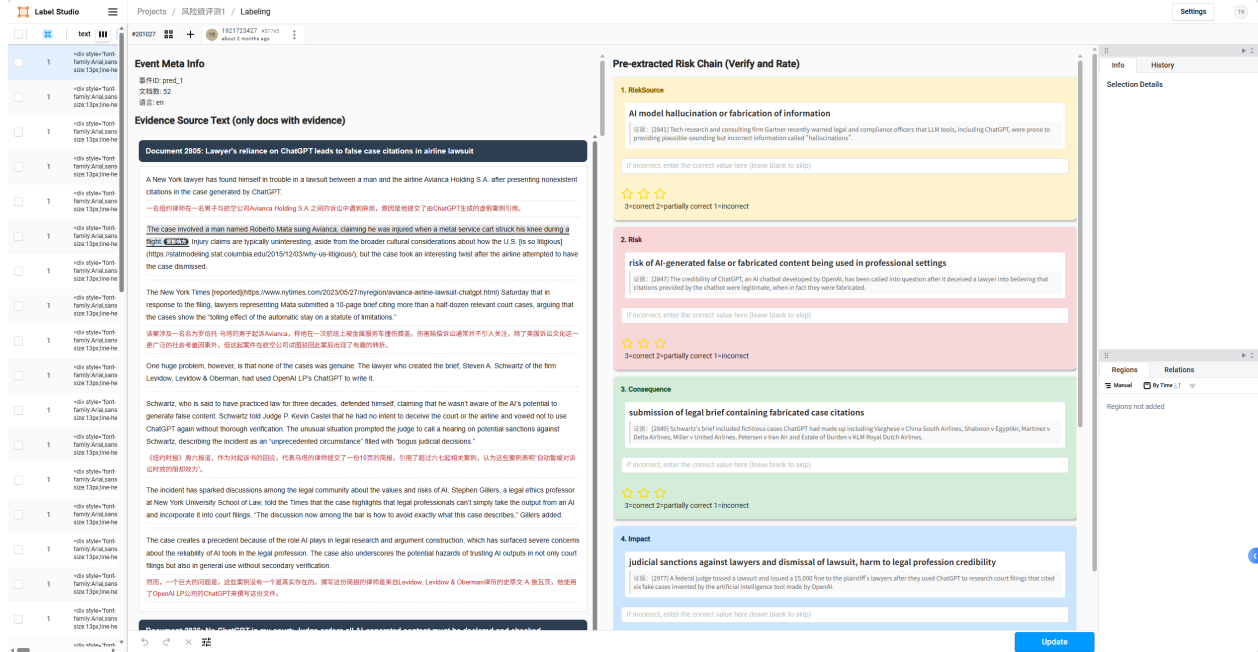


Figure C.2: Layer 2 annotation interface for risk propagation chain verification.

## C.3. Layer 3: Ontology-Guided Inference Annotation

The third annotation stage evaluates inferred knowledge generated through ontology-guided reasoning. Unlike the previous layers, inferred statements are not explicitly stated in the source documents but are derived from ontology constraints, semantic relations, and event context.

Annotators examine generated inference candidates together with supporting evidence and determine whether the inferred knowledge is logically justified. This stage is used to evaluate the reliability and usefulness of ontology-guided knowledge augmentation.

## OntoRisk: AI Risk Knowledge Graph Construction

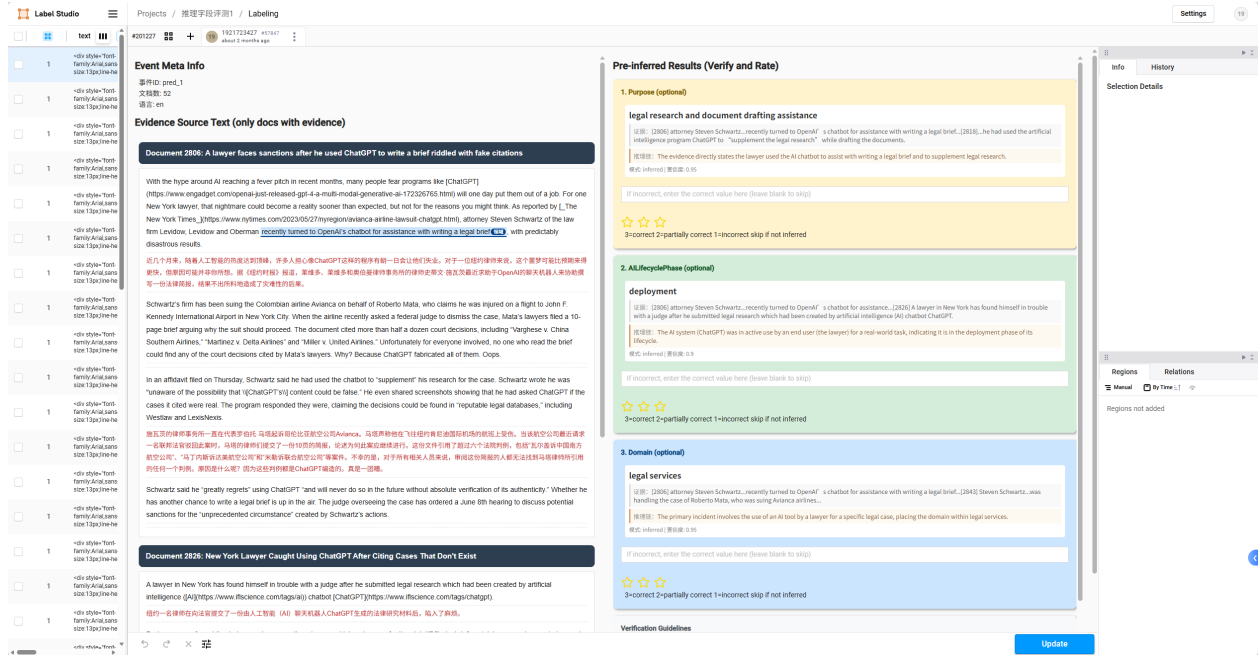


Figure C.3: Layer 3 annotation interface for ontology-guided inference validation.